



REVIEW ARTICLE | DECEMBER 28 2015

Atomic layer deposition on polymer fibers and fabrics for multifunctional and electronic textiles

Alexandra H. Brozena; Christopher J. Oldham; Gregory N. Parsons



J. Vac. Sci. Technol. A 34, 010801 (2016)

<https://doi.org/10.1116/1.4938104>



Articles You May Be Interested In

Photonic crystal textiles for heat insulation

J. Appl. Phys. (September 2023)

Towards the textile transistor: Assembly and characterization of an organic field effect transistor with a cylindrical geometry

Appl. Phys. Lett. (October 2006)

Exploration of microfluidic devices based on multi-filament threads and textiles: A review

Biomechanics (September 2013)

The logo for Hiden Analytical features the word 'HIDEN' in a large, bold, sans-serif font, with 'ANALYTICAL' in a smaller font below it. The background is a dark red with a pattern of small, glowing circles and lines.

Precision Quadrupole Mass Spectrometers

for Vacuum, Gas, Plasma & Surface Science

Trusted in Research
for over 40 years

Find Solutions for Your Research

REVIEW ARTICLE

Atomic layer deposition on polymer fibers and fabrics for multifunctional and electronic textiles

Alexandra H. Brozena, Christopher J. Oldham, and Gregory N. Parsons^{a)}

Department of Chemical and Biomolecular Engineering, North Carolina State University, Raleigh, North Carolina 27695-7905

(Received 8 October 2015; accepted 7 December 2015; published 28 December 2015)

Textile materials, including woven cotton, polymer knit fabrics, and synthetic nonwoven fiber mats, are being explored as low-cost, flexible, and light-weight platforms for wearable electronic sensing, communication, energy generation, and storage. The natural porosity and high surface area in textiles is also useful for new applications in environmental protection, chemical decontamination, pharmaceutical and chemical manufacturing, catalytic support, tissue regeneration, and others. These applications raise opportunities for new chemistries, chemical processes, biological coupling, and nanodevice systems that can readily combine with textile manufacturing to create new “multifunctional” fabrics. Atomic layer deposition (ALD) has a unique ability to form highly uniform and conformal thin films at low processing temperature on nonuniform high aspect ratio surfaces. Recent research shows how ALD can coat, modify, and otherwise improve polymer fibers and textiles by incorporating new materials for viable electronic and other multifunctional capabilities. This article provides a current overview of the understanding of ALD coating and modification of textiles, including current capabilities and outstanding problems, with the goal of providing a starting point for further research and advances in this field. After a brief introduction to textile materials and current textile treatment methods, the authors discuss unique properties of ALD-coated textiles, followed by a review of recent electronic and multifunctional textiles that use ALD coatings either as direct functional components or as critical nucleation layers for active materials integration. The article concludes with possible future directions for ALD on textiles, including the challenges in materials, manufacturing, and manufacturing integration that must be overcome for ALD to reach its full potential in electronic and other emerging multifunctional textile systems.

© 2015 American Vacuum Society. [<http://dx.doi.org/10.1116/1.4938104>]

I. INTRODUCTION

Textiles and fibers make up many of our most commonly used products. Garments provide protection, warmth, visual appeal, identification, and other benefits. Fibrous materials also appear as furnishings, wall and floor coverings, hygiene and healthcare materials, filters, cable supports, and other basic infrastructure. Textiles and fibers appear in these applications because they are renewable and low cost materials, light weight and flexible, and feature well developed manufacturing. The high surface area of textiles also provides a large active region for interaction with the environment or other target areas, including the human body. For these reasons, textiles provide a convenient and common platform to add new functionality to current products.

Often, textiles are chosen to provide more than one function in a given application. Natural wool, for example, provides physical protection and repels water while retaining its thermal insulation under both wet and dry conditions. “Multifunctional textiles” describe materials that are technologically modified to provide utility that is typically

nonaesthetic and beyond the range of common fabrics. For example, electronic coupling into textiles began as early as the development of electric blankets nearly 100 years ago.¹ Now electronic fabrics, or “e-textiles,” aim to add more function, such as communication and computational capacity into a wearable platform. Other names that describe multifunctional nonaesthetic textiles include “technical textiles” and “smart textile” systems.^{2–6}

Electronic textiles often use woven and nonwoven fiber mats as a platform to add on electronic logic, memory, display, sensing, and energy generation/storage capabilities.^{7–14} Other systems directly integrate or build in electronic functionality into individual fibers within the textile.^{14–24} Some examples of “add-on” and “built-in” electronic textiles are shown in Fig. 1. While much attention is given to e-textiles for wearable electronic systems, advanced textiles are also impacting broad new areas such as construction, environmental protection, chemical decontamination, pharmaceutical and chemical manufacturing, catalytic support, tissue regeneration, prosthetics, and other healthcare materials.^{25–34} To achieve these advances, innovations in chemistry, biological coupling, and nanodevice systems are needed that can readily combine with textile manufacturing.

^{a)}Electronic mail: gnp@ncsu.edu

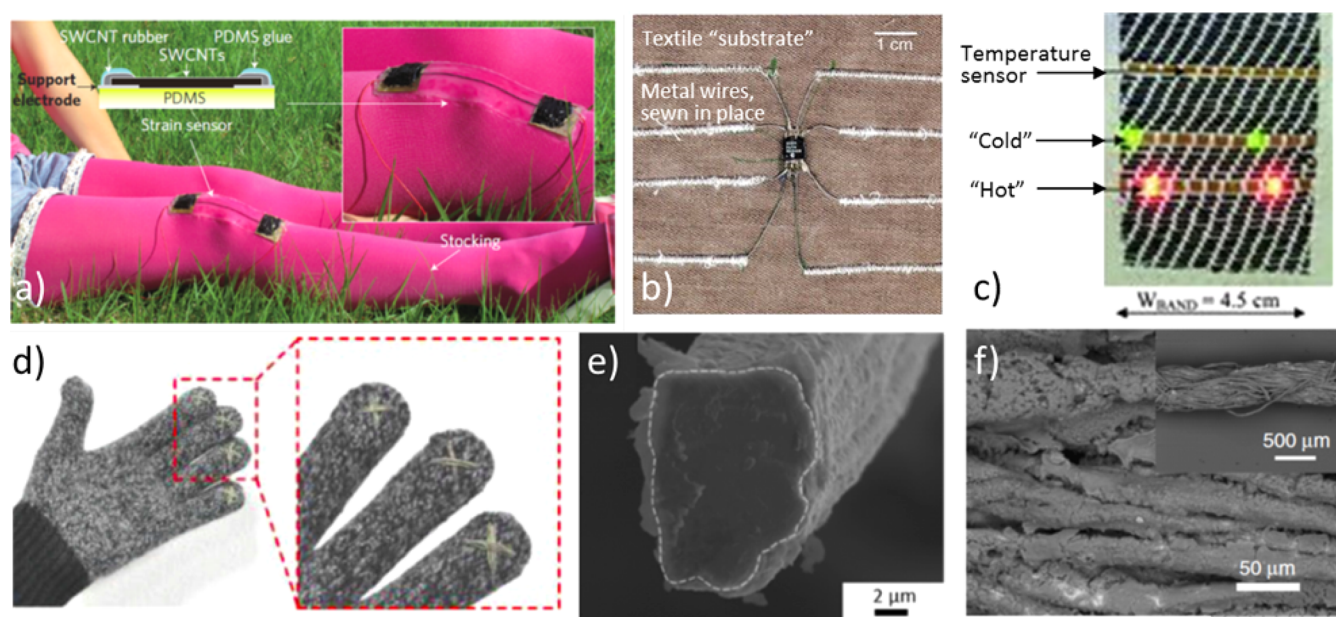


FIG. 1. (Color online) Examples of “add-on” (top row) and “built-in” (bottom row) electronically active textiles: (a) a carbon nanotube strain sensor on legging-wear; (b) a sewn-on integrated circuit; (c) flexible strips of sensors and light-emitting indicators woven into fabric; (d) individual fibers with pressure sensing capability integrated into a smart glove; (e) a weavable supercapacitor fiber using stainless steel coated with polypyrrole, MnO_2 , and reduced graphene oxide; (f) supercapacitor yarns created from cotton coated with electroless Ni and reduced graphene oxide. Sources: (a) Reprinted with permission from Yamada *et al.*, *Nat. Nanotechnol.* **6**, 296 (2011). Copyright 2011, Nature Publishing Group. (b) Adapted from Post *et al.*, *IBM Syst. J.* **39**, 840 (2000). (c) Reprinted with permission from Cherenack *et al.*, *Adv. Mater.* **22**, 5178 (2010). Copyright 2010, John Wiley and Sons. (d) Reprinted with permission from Lee *et al.*, *Adv. Mater.* **27**, 2433 (2015). Copyright 2015, John Wiley and Sons. (e) Reprinted with permission from Huang *et al.*, *ACS Nano* **9**, 4766 (2015). Copyright 2015, American Chemical Society. (f) Reprinted with permission from Liu *et al.*, *Nat. Commun.* **6**, 7260 (2015). Copyright 2015, Nature Publishing Group.

One of the most well-known attributes of atomic layer deposition (ALD) is its ability to create films and coatings with excellent conformality across large areas with complex roughness and topography. Since fibers and textiles provide unique surface structures, many teams have studied ALD as a way to modify, encapsulate, mimic, or otherwise coat fibers and textiles to add new multifunctional capabilities.^{35–40} Significant fundamental work has been done to understand the mechanisms in ALD modification of polymers^{41,42} and other fiber materials,^{43–47} and to identify and explore functionality that ALD can uniquely add to create practical textiles.

The aim of this article is to provide an overview of the current understanding and capability for ALD coating and modification of textiles, with specific attention for electronic and other emerging multifunctional textile systems. Section II provides a brief description of textile materials and processing, and discusses common research approaches for textile coating and modification, followed in Sec. III by a basic introduction to ALD on fibers including a discussion of materials and process challenges and limitations. Section IV then reviews the unique properties of ALD-coated textiles, and how ALD-based methods can modify and improve fibrous materials. In Sec. V, we show recent examples of electronic and multifunctional textiles that use ALD coatings either as functional components or as critical building-blocks for active materials integration. Finally, Sec. VI presents possible future directions for ALD on textiles and describes the challenges that must be overcome for ALD to reach its full potential in this field.

II. TEXTILE AND NONWOVEN MATERIALS FABRICATION AND MODIFICATION

A. Textile materials fabrication

Textiles, including wovens, knits, and other forms of cloth, are made of both natural and synthetic materials that are spun, extruded, or otherwise formed into fibers and yarns. Nonwovens, including spun-bond, meltblown, and electrospun materials,^{48–50} are typically comprised of small fibers with diameters of $\sim 5\text{--}20\ \mu\text{m}$ (or in the 100 nm range for electrospun nanofibers),^{51,52} which are laid down on a moving platform to form continuous fiber mats. The physical arrangement and entanglement of fibers in nonwovens can be similar to felt wool or other natural mats. Nonwoven processing often makes use of recycled plastics and other goods to produce low-cost materials with positive environmental impact. The terms “textile” and “nonwoven” technically refer to different materials with distinct properties and function. However, in this article, the terms “fabric,” textile, “fiber mat,” and others will be used interchangeably with the understanding that the resulting function or use will likely differ.

B. Textile surface modification

In textile processing, fibers, yarns, and fabrics may undergo many additional steps including dyeing, printing, and other chemical or physical “finishing” treatments.^{53,54} Finishing technology is often critical because it gives the material a desirable appearance or feel (i.e., by modifying fiber surface roughness), or a unique added function, such as

for UV protection or control of surface wetting, among other possibilities. Many of these finishing steps require heating for extended periods of time to dry the treated fabrics; so, these processes are typically energy intensive and expensive. Therefore, industry seeks new approaches that improve current function while minimizing energy use and environmental impact. As a surface modification method, ALD typically requires elevated temperatures as well, but heating times are generally shorter than traditional finishing treatments, which could result in overall lower energy use and greater economic savings.

Most common textile finishes are applied using solution-based processing. For example, cotton is made wrinkle-free with a standard “pad-dry-cure” process that uses an organic diamine crosslinker solution in the presence of an acid catalyst.⁵³ Other solution methods studied for fiber surface modification include sol-gel processing, dip-coating, electrostatic layer-by-layer deposition, spray coating, direct chemical grafting, and others.^{2,55–63} Uniformity in solution-based processes is often acceptable, but not ideal. Environmental compatibility is another growing concern.

Vapor-phase processing is also used for surface finishing in textile manufacturing. Plasma surface modification can be done using inert or chemically active gases, and processes can be scaled to roll-to-roll at atmospheric pressure.^{64,65} Acid groups, for example, can be attached to fibers via direct plasma excitation of the acid vapor, or they can be attached via solution processes after treatment by inert gas plasma to create functional radical sites on the fiber surface.⁶⁵ The plasma modification approach can work well to modify the surface-face of the textile, but modification is generally non-uniform throughout the fiber matrix.⁶⁶ Plasma modification requires excited species (e.g., radicals and/or neutral excited metastable atoms) to interact with the polymer, but these excited species readily degrade or recombine to their ground state upon collision with other gases or with the polymer surface. The degradation rate depends on the gas mean-free-path, the distance between polymer fibers, and the net fiber mat thickness. Therefore, getting radicals to penetrate beyond the near surface-face of the fiber mat is difficult, resulting in nonuniform modification of the fibers within the mat. For these same reasons, plasma modified ALD is limited in terms of uniform deposition of surface coatings on textiles and fiber mats. ALD and non-ALD treatments using other excited species that degrade upon collision, including ozone for example,⁶⁷ are also expected to have similar uniformity problems. Nevertheless, vapor modification by plasma or excited species can be a viable approach for many chemical treatments.

In addition to plasma techniques, several other vapor-based methods have been evaluated for textile treatment, including physical, chemical, and plasma vapor deposition; vapor chemical grafting; and vapor-sprayed electrolyte assemblies.^{60,61,68–72} Early studies of fiber surface modification explored UV-promoted vapor chemical grafting.⁷² Armstrong and Rutherford⁷² showed that under UV irradiation, vinyl-containing acrylate, vinyl acetate, and styrene monomer vapors would readily react and bind to celluloses,

cellulose esters, polyamide, and polypropylene. They found that grafting a small amount of acrylonitrile onto cotton made it highly resistive to penicillium oxalicum, a cellulose-destroying fungus. They also found that grafting reactions were not limited to the fiber surface. UV grafting of polyacrylonitrile to acetate fibers increased the Young’s modulus, which the authors ascribed to subsurface monomer diffusion and crosslinking. However, the UV grafting procedure lacked uniformity. Loosely wound yarns could be coated evenly, whereas uniformity was poor on fiber mats.⁷²

Direct chemical vapor deposition (CVD) typically requires high temperatures which are not suitable for many polymer fibers. However, using a vapor-phase oxidant or radical initiator, oxidative and initiated-CVD (*i*-CVD) can produce polymer coatings at low temperature.⁷³ For example, Ma *et al.* used *i*-CVD of perfluoroalkyl methyl methacrylate with a *tert*-butyl peroxide initiator to create a superhydrophobic coating on electrospun fiber mats.⁶⁸ In this process, the extent of polymerization was determined by the availability of the active initiator, and conformal polymer coatings were achieved when sufficient initiator penetrated through the fiber mat. In thicker mats, a profile of initiator surface concentration would be expected to lead to thicker polymer growth on top of the exposed mat surface.

New surface treatments are needed for future multifunctional or smart textiles, especially those that directly couple new electronic, sensing, chemical, security, and other functionalities. In addition, new surface modification strategies that modify surface wetting, add chemical stability, control thermal conductivity, allow selective adhesion, or adjust biological response may prove to be more important as a means to prepare fibers and textiles for subsequent processes that would otherwise not be viable on fibers. Such processes could promote new low-cost multifunctional fiber and fabric materials for high-surface area catalysis, large-area chemical remediation, rapid low-energy water purification, and other areas of need.

III. ALD ON TEXTILES AND FIBERS

ALD proceeds using alternating vapor exposures under conditions that create a sequence of self-limiting surface chemisorption reactions.^{74–76} The process is well known and widely used in large-scale semiconductor manufacturing, in part, because it can deposit inorganic thin films with high conformality and precise nanoscale thickness on complex high-aspect ratio geometries. The self-limiting nature of ALD surface reactions allows uniform coatings on nonplanar, porous, and otherwise tortuous substrates. Early work recognized that fiber substrates are a unique platform for ALD to create new three-dimensional inorganic lattices with chemical functionality (Fig. 2).^{43,77}

A. Basics of ALD for fiber and textile applications

The ability of ALD to coat complex fibrous surfaces in a conformal manner is due to its self-limiting reaction chemistry. ALD typically uses two half-reactions which proceed by alternately pulsing each reaction precursor onto the

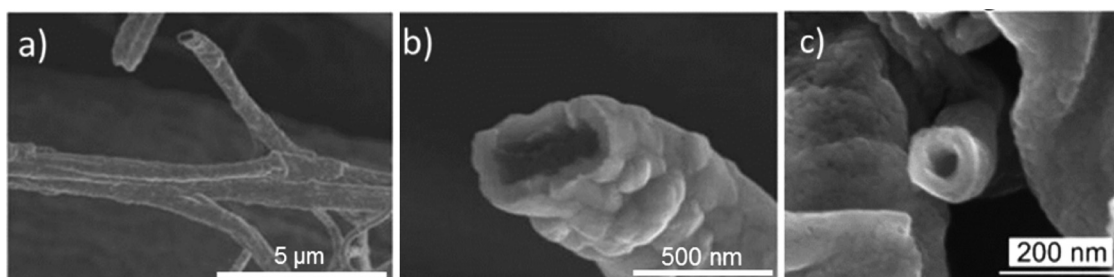


FIG. 2. TiO_2 coatings deposited on cellulose fibers using ALD. The paper fibers were removed by annealing in air, leaving a TiO_2 shell that replicated the original shape of the substrate. (a) and (b) Reprinted with permission from Ritala *et al.*, Chem. Vap. Deposition **12**, 655 (2006). Copyright 2006, John Wiley and Sons. (c) Reprinted with permission from Kemell *et al.*, J. Am. Chem. Soc. **127**, 14178 (2005). Copyright 2005, American Chemical Society.

substrate's surface in a batch reactor system. Robust ALD processes employ half-reactions that are each exothermic, helping to promote low temperature growth. Following each exposure step, excess precursor and reaction byproducts are removed with inert gas flow. Insufficient purging can lead to gas phase reactions that lead to nonuniform and/or CVD-like growth.

ALD on polymers often features precursor and reactant diffusion into the near-surface or bulk of the polymer.⁴² Figure 3 shows two schematic diagrams for Al_2O_3 ALD growth on polymer fibers.⁷⁸ In Fig. 3(a), the cotton fiber substrate contains sufficient reactive groups to rapidly initiate growth. The trimethyl aluminum (TMA) exposure creates terminal aluminum methyl bonds on the fiber surface that react and oxidize during subsequent water exposure. The figure includes a cross-sectional transmission electron microscopy (TEM) image of a cotton fiber after Al_2O_3 ALD, showing a relatively abrupt organic/inorganic interface. The abundance of Lewis basic sites, such as hydroxyl groups on the surface of many natural fibers allows ALD reactions to begin efficiently by using Lewis acidic reactants, including TMA, titanium tetrachloride (TiCl_4), and others. Natural fibers, such as wool, cotton, and other cellulose derivatives, also make interesting substrates for ALD modification due to their renewable, inexpensive, high surface area, and mechanically resilient nature. For example, paper, made of compressed cellulosic fibers and nanofibrils, has been modified using ALD with repeated success.^{79–82}

Different polymers and different ALD reactants can produce a very different outcome, as shown in Fig. 3(b). In this system, reactants diffuse into the polymer leading to nuclei formation within the near-surface region, swelling the fiber, and creating a roughened surface texture. A cross-section of the polypropylene fiber coated using TMA/water at 90°C shows an example outcome. However, as the nuclei continue to grow, the polymer becomes covered with metal oxide, allowing uniform ALD to eventually proceed.^{35,42,83} This process can be strongly temperature dependent. Jur *et al.* found that when the deposition temperature for Al_2O_3 ALD on polypropylene was reduced to 60°C , an abrupt interface formed between the polymer and the metal oxide coating, indicative of more uniform Al_2O_3 growth on the fiber surface as determined by TEM.³⁵ Understanding these kinds of ALD growth mechanisms on polymers and fibers allows

researchers to design more effective reactions to achieve the desired fiber coating with targeted functionality. For example, Sweet *et al.*⁸⁴ showed that a uniform conductive ZnO ALD layer could be formed on polypropylene at 155°C by first depositing a uniform base-layer of low temperature ($\sim 60^\circ\text{C}$) ALD Al_2O_3 . The Al_2O_3 pretreatment helped prevent subsurface precursor diffusion during the higher temperature ZnO ALD, thereby enabling conductivity of $>60\text{ S/cm}$, as compared to just 15 S/cm without the base layer.

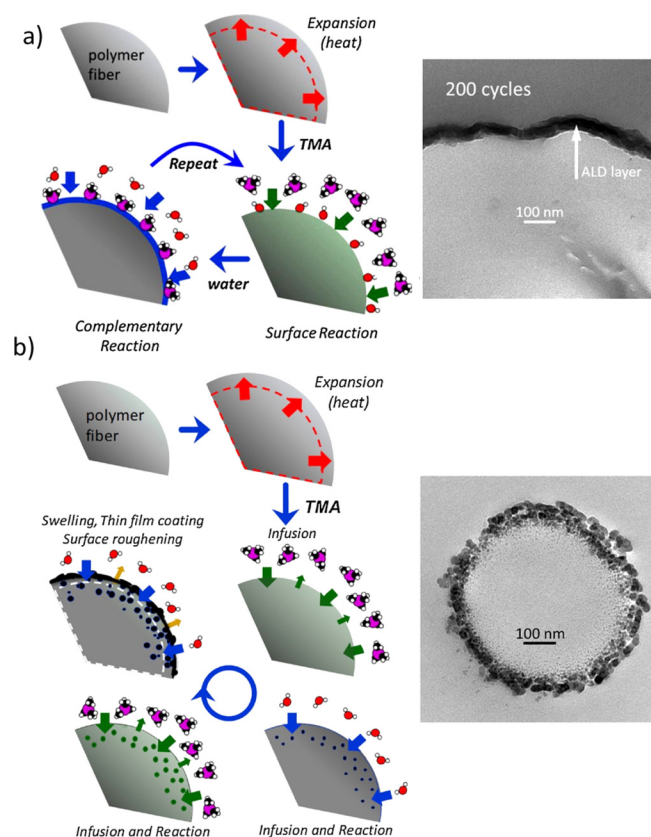


FIG. 3. (Color online) (a) ALD on a hydroxyl rich polymer surface leads to conformal deposition as shown in the example of Al_2O_3 ALD growth on cotton (top TEM image). (b) ALD on a polymer surface that lacks reactive functional groups can result in subsurface diffusion and growth within the fiber core, resulting in a less uniform coating at some temperatures. ALD growth on polypropylene at high temperatures (90°C) results in this kind of subsurface deposition (lower TEM image). Reprinted with permission from Parsons, *Atomic Layer Deposition of Nanostructured Materials*, edited by N. Pinna and M. Knez (Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim, Germany, 2011), pp. 271–300.

Under certain conditions, the transport of reactants can be very rapid in some polymer materials and negligible in others. Gong *et al.*⁸⁵ demonstrated selective reactant transport and reaction by treating bicomponent fibers with ALD. When fibers with a polylactic acid core and polypropylene sheath were exposed to short TMA/water dose steps at 120 °C, TEM analysis showed a thin conformal metal oxide on the outer polypropylene surface. However, when longer exposures were used, TMA was found to diffuse through the polypropylene and selectively react with the central polylactic acid core region. Selective polymer modification via ALD could extend to other multicomponent materials to create new polymer/inorganic nanocomposite textiles. In subsequent experiments, the extent of subsurface diffusion was also found to depend on reactant purge time, where longer purge steps allowed more precursor to be transported out of the polymer bulk.

These examples show possible outcomes during ALD on polymers, but other processes could also occur, including chemical and structural modification of the polymer itself. It is important to note that even under conditions that lead to well-controlled ALD on solid surfaces, reactions can occur on or within polymers that are beyond those described as ALD.

B. Temperature limitations of textile materials

A wide variety of natural and synthetic materials are used in woven and nonwoven textiles. Table I presents some common textile fiber materials, along with the typical maximum process temperature that the material can withstand during short processing time. The overall thermal budget must be considered when utilizing higher temperatures, so extended heating should be done below the maximum process temperature. Most ALD chemistries proceed best within a targeted “ALD temperature window” where full surface saturation occurs during each reaction half-cycle. For several metal oxides, such as Al₂O₃, ZnO, and TiO₂, thermal ALD processes are well characterized at temperatures compatible with textiles. However, for thermal ALD of elemental metals, higher temperatures are typically needed to achieve full metal reduction or ligand combustion. Some metals such as

TABLE I. Maximum process and glass transition temperatures of several common polymers found in fibers and textiles, however, under extended heating, process temperatures below these maximum values are preferred to maintain the quality of the substrate.

Material	Maximum temperature (°C)	Glass transition temperature (°C)	References
Cotton	330–360		86
Nylon-6	220	50	87
Polypropylene	165	–10	88,89
Polyethylene terephthalate	250–270	67	90–92
Polybutylene terephthalate	230	120	91,93
Polyethylene naphthalate	275	120–135	92
Kevlar®	540		94
Polyvinyl alcohol	220–230	85	95,96
Polyacrylonitrile	240	95	97,98
Poly(vinylidene fluoride)	160	–35	99,100

Cu and W can be formed by thermal ALD at less than 230 °C. W ALD from SiH₄ and WF₆ has been studied, in particular.¹⁰¹ Plasma or other radically enhanced ALD processes have been studied at low temperatures on textiles,^{45,102–104} but these kinds of reactive ALD processes do not achieve full reactant penetration into thicker fiber mats, thereby limiting deposition to the surface-exposed fibers. This ability to limit the growth to the top of fiber surfaces may prove useful for some applications.

C. ALD conformality and uniformity within fiber mats

In principle, the self-limiting half-reactions in ALD allow a uniform and conformal coating to form on all surfaces exposed in the reactor deposition zone. However, full growth requires reactant saturation, which in turn requires each local surface to receive sufficient vapor exposure during each half cycle. Reactant purge steps between each half cycle must also be sufficient to remove byproducts and unused reactants.

Most studies of ALD on textiles use viscous-flow batch ALD reactors where the reactants flow over the fabric sample. In this configuration, reactant exposure time must be sufficient to allow species to diffuse into the regions between fibers. Likewise, purge cycles must be long enough to allow reaction products to desorb and diffuse out of the fiber structure and into the reaction flow stream. In this case, the mean-free path for gas phase species within the fiber mat can be close to the fiber-fiber distance, leading to relatively slow Knudsen or molecular flow conditions.

To modify reactant and byproduct transport in porous fibrous materials, Ritala *et al.*⁷⁷ demonstrated a “flow-through” reactor design where the reactant species were convectively forced to flow between fibers within a filter paper substrate. The pores size allowed sufficient gas conductance to avoid a large pressure drop across the substrate. Hyde *et al.*¹⁰⁵ also explored the issue of flow geometry during ALD of Al₂O₃ on polypropylene [Fig. 4(a)] and found that orienting nonwoven polypropylene mats parallel or perpendicular to the gas-flow, i.e., “flow over” or “flow through” configurations, produced similar coatings under the conditions used.

Many studies^{35,43,81,101,105–109} have confirmed the conformality of ALD coatings on fibers using electron microscopy analysis. Based on the analysis of the fiber materials and ALD process conditions in these reports, we find that when fiber mats with moderate density (i.e., open volume fraction of 50%–85%)¹⁰⁶ are coated under ALD conditions using moderate reactant exposure (e.g., ~1 s/cycle at 1–10 Torr), the fibers pulled from the fiber mats generally show good uniformity and excellent conformality. The overall surface area of the fibers can be quantified in terms of a “surface area enhancement factor,”¹⁰⁶ or aspect ratio,^{45,110,111} i.e., the total surface area of the fibers per unit projected surface area of the fiber mat. This ratio is effectively equivalent to the aspect ratio used for trenches, vias, and straight pores. Although this aspect ratio is not directly correlated with open volume fraction, nonwoven microfiber mats with open volume fraction of 75%–90% have an effective aspect ratio of 50–200:1, depending on the fiber diameter.¹⁰⁶ This aspect

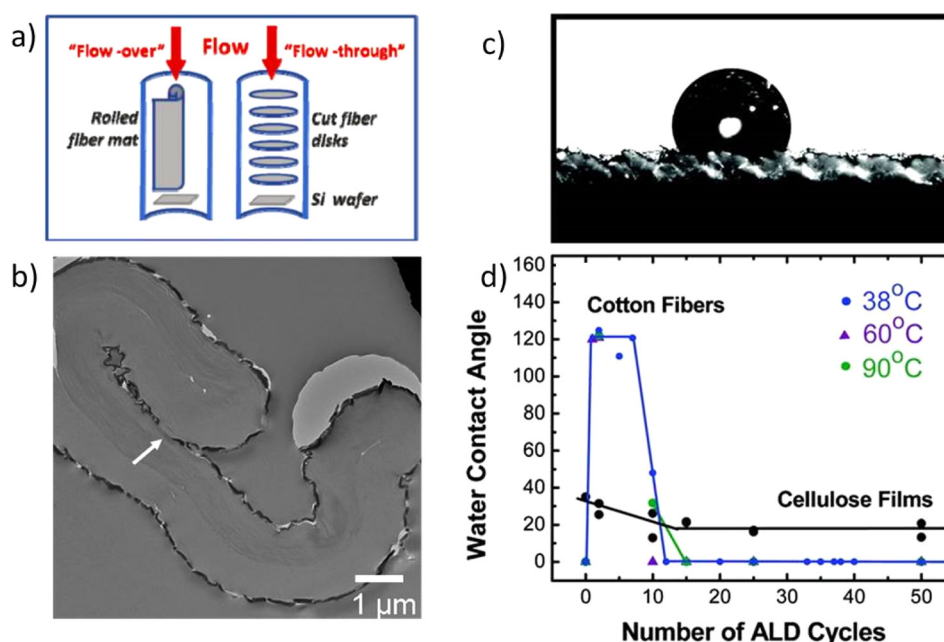


FIG. 4. (Color online) a) Flow-over and flow-through configurations for ALD on fibers. (b) A cross-sectional TEM image of a cotton fiber showing uniform Al_2O_3 ALD on the complex bean shape. Reprinted with permission from Hyde *et al.*, *Langmuir* **26**, 2550 (2010). Copyright 2010, American Chemical Society. (c) Water droplet on hydrophobic cotton after a few ALD cycles. (d) Water contact angle on cotton fibers and cellulose films after ALD coating. Reprinted with permission from Hyde *et al.*, *Langmuir* **23**, 9844 (2007). Copyright 2007, American Chemical Society.

ratio is sufficiently small that ALD results in uniform and conformal coatings with precise thickness control on individual fibers throughout the fiber network.

An example of an ALD Al_2O_3 coating on a common cotton fiber with a “bean” cross-sectional shape is shown in Fig. 4(b). In this case, the uniformity of the coating on the cotton surface indicates consistent saturation during each ALD reaction half-cycle. It is important to note that the thickness of deposited inorganic coatings on polymer fibers is difficult to quantify by TEM using samples prepared by microtome sectioning. A typical microtome slice is ~ 100 nm thick. When the microtome cuts a hard coating on a softer polymer, the coating can twist, fold, or crack, distorting the thickness analysis. Visible faults in the ALD layer in Fig. 4(b) are ascribed to cracking during microtome preparation.

Ideally, one would like to quantify coating uniformity through the fiber mat. In one report, infrared (IR) transmission spectroscopy was used to assess uniformity of ALD coatings within nonwoven mats by analyzing fiber sections.¹⁰⁵ Nonwoven mats are often made of multiple layers of overlaid fibers, and the layers can be mechanically separated or peeled from each other. Following Al_2O_3 ALD under typical moderate conditions, nonwoven fiber mats were physically peeled into thin layers and measured using IR spectroscopy to monitor the intensity of the Al–O stretching peak. Normalizing the intensity using characteristic IR peaks in the polymer, the Al–O intensity was found to be the same on each of the five layers extracted from a single nonwoven mat, demonstrating uniform coating throughout the mat using practical ALD conditions.

A more full understanding of ALD reactant transport and surface saturation within and through complex surface geometries can be approached through experiment and modeling.^{45,105,106,112,113} Musschoot *et al.* studied the uniformity

of ALD Al_2O_3 on nonwoven polyester fiber mats using scanning electron microscopy (SEM) and energy-dispersive x-ray analysis (EDX).⁴⁵ By allowing reactants to diffuse into the nonwoven mat on only one face, the authors measured the deposition penetration distance into various mats versus reactant exposure time and compared results to a random diffusion and reaction model. Under ALD conditions, a few seconds of reactant exposure per cycle produced a coating extending 5–10 mm into the fiber mat. The extent of coating and coating profile were consistent with model predictions. Furthermore, the penetration distance decreased significantly using plasma ALD. Specifically, while thermal Al_2O_3 ALD penetrated more than 10 mm into the mat, replacing water with plasma-excited oxygen led to less than 3 mm penetration, consistent with rapid decay of plasma excited species via collisions in the fiber network.

In another study, Sweet *et al.* used local compression of nonwovens to adjust and analyze diethyl zinc (DEZ) reactant penetration through fiber networks during ALD of ZnO. Growth profiles measured with the fiber mat agreed with model predictions, where a few seconds of DEZ exposure per cycle coated the fibers within 1–2 mm of the top mat face.¹⁰⁶ The deeper penetration observed by Musschoot *et al.*⁴⁵ was consistent with the larger void fraction in their nonwovens ($\sim 95\%$) compared to those materials studied by Sweet *et al.* (50%–80%).¹⁰⁶

IV. PROPERTIES OF ALD COATED TEXTILES AND FIBERS

A. ALD for control of surface energy and wetting

The demand for self-cleaning and high-performance textiles provides ample motivation to study how ALD can

control surface wetting properties of fibers. The nature of water interaction and wetting on textured surfaces depends on both the fractional surface area in contact with the water and the surface energy of the material. Increasing the surface energy, such as through the deposition of polar metal oxide nanocoatings using ALD, is one way to increase the wetting capabilities of the substrate.^{114–116} Upon coating by Al_2O_3 ALD, nonwetting hydrophobic polypropylene fibers can transition to either a metastable hydrophobic or a fully wetting hydrophilic state, due in part to the natural wettability of the polar Al_2O_3 surface.¹⁰⁵ On the other hand, when naturally wetting fabrics such as cotton are coated by ALD metal oxide, a few ALD cycles can cause a transition to a metastable hydrophobic state [Fig. 4(c)] and back to wetting depending on the number of ALD cycles [Fig. 4(d)].^{105,117} The transition between wetting and nonwetting occurs abruptly with the number of ALD cycles. Correlating the surface wetting characteristics of coated planar and textured fabric surfaces, the observed trends can be understood in terms of the well-known Cassie-Baxter and Wenzel models for wetting of roughened and textured surfaces.^{118,119}

The wetting transitions on polypropylene are found to proceed more readily when Al_2O_3 ALD is done at low temperature (e.g., 60°C), whereas coatings at higher temperature ($>90^\circ\text{C}$) often retained the hydrophobic state. Subsequent studies concluded that surface roughening caused by excess subsurface reactant penetration at higher temperatures likely helped the fibers to remain nonwetting. In contrast, the wetting transitions on cotton are less sensitive to deposition temperature. IR and x-ray photoelectron spectroscopy analysis suggests that the rapid hydrophobic transition occurs when TMA consumes hydroxyl groups exposed on the cotton cellulose surface to create hydrophobic $\text{Al}(\text{OC})_3$ surface groups. Some surface roughening was also observed, which may also promote the hydrophobic transition. The hydrophobic state is relatively stable in air, but it can degrade over time upon coating by adventitious carbon. Deposition greater than ten cycles eventually overwhelms these effects, resulting in a smoother, $\text{Al}-\text{OH}$ terminated surface, which raises the surface energy of the treated cotton, causing it to become hydrophilic once more. Similar trends were observed for ZnO deposition as well.¹¹⁷ Alternative coatings, such as TiN , can also increase the hydrophobicity of cotton fabric.¹²⁰ The change in surface wetting of fiber mats using an ALD coating can be an important way to enable subsequent wet processing.

B. Conductive coatings on textiles

Conductive fibers are attractive for many smart textile applications, including embedded sensors, flexible antennas, and displays.¹²¹ Depositing metal in a conformal manner and with controllable thickness is essential to give flexible, fibrous materials uniform conductivity that is comparable to the bulk coating material. Solution methods are generally used to add conductivity to fabrics,^{2,55–63} but they often add unwanted weight, change the feel, or reduce fabric flexibility. Conductive carbon nanotubes, for example, can be added to

create composite fibers, but conductivity is often less than 10 S cm^{-1} .¹²² Likewise, Kevlar fibers coated with single-walled carbon nanotubes has conductivity limited to 65 S cm^{-1} .¹²³

While many metal ALD processes are known, most often these require reduction or combustion steps at 250°C or above. Most studies of conductive ALD coatings on textile fibers focus on lower temperature thermal or plasma ALD materials, including W, Cu, Al-doped ZnO , ITO, and Pt.^{79,84,101,104,106,124–126} One challenge in metal ALD on fibers is to quantify the resulting conductivity of the deposited material. To address this issue, Jur *et al.*⁷⁹ described a modified four-point probe to quantify fiber coating conductivity. The conductive percolation between fibers was maximized by physical compression, and the conductivity was estimated using the volume and known density of the deposited thin film.¹²⁵ Using this characterization method, ALD W films deposited at 200°C on quartz fibers were found to have a conductivity of $\sim 1150\text{ S cm}^{-1}$, consistent with that of other W thin films, but less than the bulk W conductivity.⁷⁹

C. Mechanical modification via vapor infiltration and metalation

Although subsurface ALD reactions can occur, coalescence of the thin film into a continuous coating will ultimately block diffusion of the reactants into the fiber. However, there are several cases where precursor diffusion and subsurface reaction may be desired to modify and improve the bulk properties of the fiber.^{47,127–130} Bulk-infiltration of reactants (typically ALD metal halides or metal organic precursors) occurs during long exposure times, sometimes followed by relatively short coreactant exposure. This results in metalation of the organic compound, modification of the polymer crystals, and/or crosslinking between neighboring chains, sometimes leading to significant changes in the mechanical properties of the fibers. For example, Gregorczyk *et al.*¹²⁸ observed that for a limited number of DEZ infusion cycles (<4), the ultimate tensile stress of cellulose fibers nearly doubled in value while the elastic modulus increased by nearly $2.5\times$. The authors ascribed this change to $\text{Zn}-\text{O}$ covalent crosslinkages between the cellulose units. They also saw that increased cycles of infusion or reaction with the more acidic TMA molecule caused the mechanical behavior to regress, possibly due to degradation of the polymer backbone (Fig. 5). Similarly, Seung-Mo *et al.*⁴⁷ observed that applying ALD coatings using DEZ, titanium isopropoxide, and TMA to spider silk increased the fiber toughness, where the extent of change increased with multiple doses of the precursor without water exposure.

D. Biocompatible and antimicrobial coatings

Understanding surface biocompatibility and controlling antimicrobial function are important challenges in surface engineering. Textiles with antiodor or antibacterial properties are used in athletic and medical applications, and fiber based drug-release devices are becoming more common. Fibrous materials are also used as scaffolds or supports for tissue-engineering and biomedical implants.^{32–34} Several

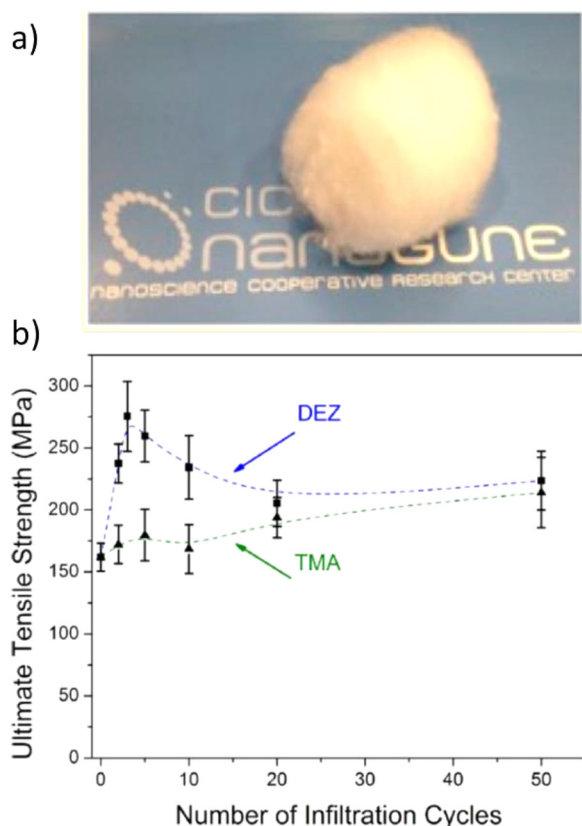


FIG. 5. (Color online) (a) Cellulose cotton fibers after vapor metalation by DEZ. (b) Tensile strength of cotton improves after metalation by DEZ, with less change after treatment by TMA. Reprinted with permission from Gegorczyk *et al.*, Chem. Mater. **27**, 181 (2015). Copyright 2015, American Chemical Society.

research groups have explored ALD for biocompatible and bioadhesive surface treatments,^{120,131,132} and as a means to modify nanomaterial toxicity.¹³³ ALD alumina¹³¹ and hydroxyapatite¹³² can also promote cell adhesion. However, the bioactive structure of hydroxyapatite needed for good cell growth requires film crystallization at 500 °C, which precludes from use on most textiles.

The superior corrosion resistance of TiN, in addition to its mechanical properties and low cytotoxicity, makes it

interesting for biomedical applications. Using tetrakis(dimethylamido)titanium and NH_3 at 100 °C, Hyde *et al.*¹²⁰ formed ALD titanium oxynitride on cotton fibers and found that human adipose-derived adult stem cells (hADSC) grew more densely on the coated versus uncoated fibers. Several thicknesses of the TiON coating were evaluated, and all were cytocompatible with good adhesion and spread of hADSC (Fig. 6).

E. Fiber encapsulation and protection

Physical stabilization and protection is another area where ALD has shown value on textiles and fibers. For example, Roy *et al.*⁴⁶ found after depositing 45 nm of ALD Al_2O_3 onto carbon fibers, the oxidation temperature increased from 250 to 600 °C, behaving more like a temperature-insensitive ceramic than an organic material. Another growing area of interest for surface modification of textiles is UV stabilization and protection. UV exposure degrades fiber mechanical performance and leads to visible color change. Hanson *et al.*⁸⁰ showed that ZnO ALD coatings on paper provided good stability against UV degradation. Moreover, the ZnO layer acted to adjust and controllably neutralize the acid content of archived paper. The good UV absorbance of many metal oxides makes ALD a viable route for UV protection, but new advances are needed to increase process speed and reduce cost.

In addition to protection from high energy sources, ALD coatings can protect textiles from mechanical forces relevant for personal protection. Kevlar®, for example, provides excellent protection from blunt impact (i.e., ballistic trauma); however, it is susceptible to cutting or slashing by sharp objects. Therefore, when both ballistic and cut threats are present, personnel must wear multiple heavy protective garments. Combining blunt force and cut protection in one material could significantly reduce the weight of protective gear. Atanasov *et al.*¹³⁴ examined various types and combinations of ALD materials to address this need. Using Al_2O_3 ALD coatings from TMA/water at 50 or 100 °C, spectroscopic and TEM analysis showed that TMA tended to react with the Kevlar® and degrade its mechanical performance.

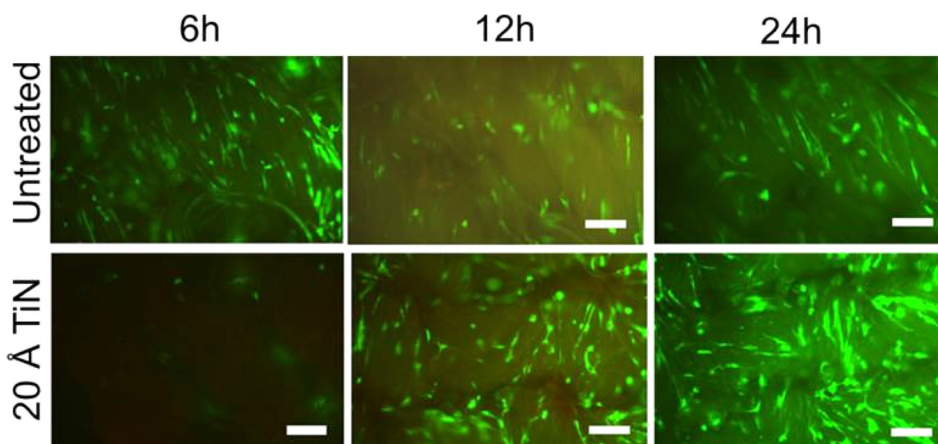


FIG. 6. (Color online) Depositing 20 Å of TiN made cotton fabric hydrophobic and helped increase the growth and spread of hADSC cells over a 24 h period. Scale bar = 200 μm . Reprinted with permission from Hyde *et al.*, Biomed. Mater. **4**, 025001 (2009). Copyright 2009, IOP Publishing.

However, TiCl_4 was more prone to react on the fiber surface and moderately improved cut performance. A 30% improvement in cut resistance was obtained after coating the Kevlar[®] fibers with 5 nm TiO_2 followed by 5 nm Al_2O_3 at 50 °C, which the authors attributed to TiO_2 forming a good interface with the Kevlar[®] to create a protective layer to isolate the material from subsequent damage during TMA exposure, and providing a good starting surface for conformal Al_2O_3 ALD. The harder top Al_2O_3 layer then helped reduce damage during cutting. The ability of ALD to improve cut resistance of Kevlar[®] is a good example of how this vapor-phase technique can add “multifunctional” performance to a textile material.

Inorganic ALD layers on polymer films can significantly reduce transmission of water and other vapors into and through the polymer.¹³⁵ However, metal oxide coatings are intrinsically stiff and they can easily crack or degrade when flexed, compromising water-resistance. One approach to make more mechanically robust protective films is to incorporate organic linking groups into the ALD sequence. For instance, organic molecules, such as glycidol or ethylene glycol, can be used in place of water as the oxidant in the reaction and alternately pulsed with ALD organometallic reagents to deposit inorganic–organic hybrid thin films in a method called, molecular layer deposition (MLD).¹³⁶ The resulting MLD processes can create fully organic polymer or oligomer films, or covalent metal oxide–organic network polymers, combining a variety of metals with a wide set of possible organic reactants that can be conformally deposited on high surface area materials like electrospun polyvinyl alcohol and nylon-6 nanofibers.^{116,136,137} These “metalcone” MLD thin films can also be combined with ALD layers to improve overall mechanical performance of oxygen or water barrier coatings on planar polymer substrates.^{138,139}

Nanofibers are attractive for biological or chemical filtration applications, but their large surface-to-volume ratio makes them inherently unstable under exposure to organic solvents. Various surface modification and treatments, including ALD, have been used to add functionality or improve stability.^{37,116,140,141} However, while some nanofibers can be readily encased by ALD metal oxides, the ALD process can also damage or chemically degrade the nanofiber. The amide bonds in electrospun nylon nanofibers, for example, are highly reactive with TMA.¹¹⁶ The challenge of improving chemical stability of nanofibers while maintaining mechanical flexibility was addressed by Oldham *et al.*¹¹⁶ using a combination of ALD and MLD. The authors found that a thin ALD layer of ZnO could encase the fiber, allowing a mechanically flexible MLD layer to be subsequently deposited using TMA/glycidol. The bilayer coating enables the fiber mat to retain its nanofiber structure under chemical exposure, while maintaining sufficient flexibility to perform well during load-bearing air filtration (Fig. 7).

F. ALD surface priming for subsequent reaction

ALD coatings can be added to fibers to form a reactive base layer, thereby enabling subsequent functionality to be

uniformly added throughout the textile material. Peng *et al.*¹⁴² showed the basic premise of ALD-based surface chemical activation of inert polymer fibers. Using an ALD metal oxide coating on polypropylene, the authors created a high density of well-defined reactive sites (metal hydroxide groups) that allowed subsequent chemical functionalization via vapor phase grafting of aminopropyl triethoxysilane (APTES). After immersing the APTES functionalized polypropylene in Au colloidal (40 nm) solution for 24 h, the fiber matrix took on a deep red color, due to electrostatic nanoparticle interactions with the available amine sites on the APTES. The authors also noted that the uniformity and chemical versatility of vapor-phase ALD provides advantages over other methods¹⁴³ for activating fiber surfaces.

In addition to organic monolayer functionalization, ALD on fibers can be useful to create well defined nucleation layers for subsequent hydrothermal^{144–146} or layer-by-layer synthesis.¹⁴⁷ ALD ZnO on polymer fibers seeds low-temperature hydrothermal growth of ZnO nanorods, creating high surface area photocatalytic fibers.^{144,145} Recent results also show that ALD layers can nucleate high-quality metal organic framework crystals,^{146,147} leading to new options for chemically absorptive and catalytic fibers and textiles, as discussed in more detail in Sec. V.

V. EXAMPLES OF MULTIFUNCTIONAL AND ELECTRONIC TEXTILES ENABLED BY ALD

The self-limiting reactions during low-temperature ALD provides a unique ability to create uniform and conformal coatings and surface modification on fibers. Moreover, under typical laboratory ALD conditions, coating uniformity can extend throughout moderately dense fiber mats and textile fabrics. In this section, we review recent examples where ALD is used to directly add electronic or multifunctional performance to textiles.

A. Electronic conduction, capacitance, and sensing

Reliable and uniform conductivity is a primary need for electronic device integration into textiles. Several examples have shown how conductive ALD coatings can be used to form functional electronic devices on a textile platform. Jur *et al.*,⁷⁹ for example, created a functional liquid chemical flow-through capacitive sensor using fiber electrodes coated with ALD W. Two pieces of W-coated material were sewn together with an intervening layer of polypropylene to form a simple metal–insulator–metal capacitor. As an all-fabric sensor, the device showed good sensitivity to different liquid chemicals (Fig. 8). In this case, the nonwoven insulator within the capacitor served as a passive sponge to collect the liquid being sensed. In future systems, material coatings that allow selective chemical adsorption or capture could be used to create wearable or otherwise widely deployable capacitive sensing elements that can electronically detect and react to specific chemical species.

Kalanyan *et al.* also used W ALD on fiber mats for functional electronic demonstration.¹⁰¹ Nonwoven nylon-6 was precoated with 2–3 nm of Al_2O_3 ALD followed by W ALD

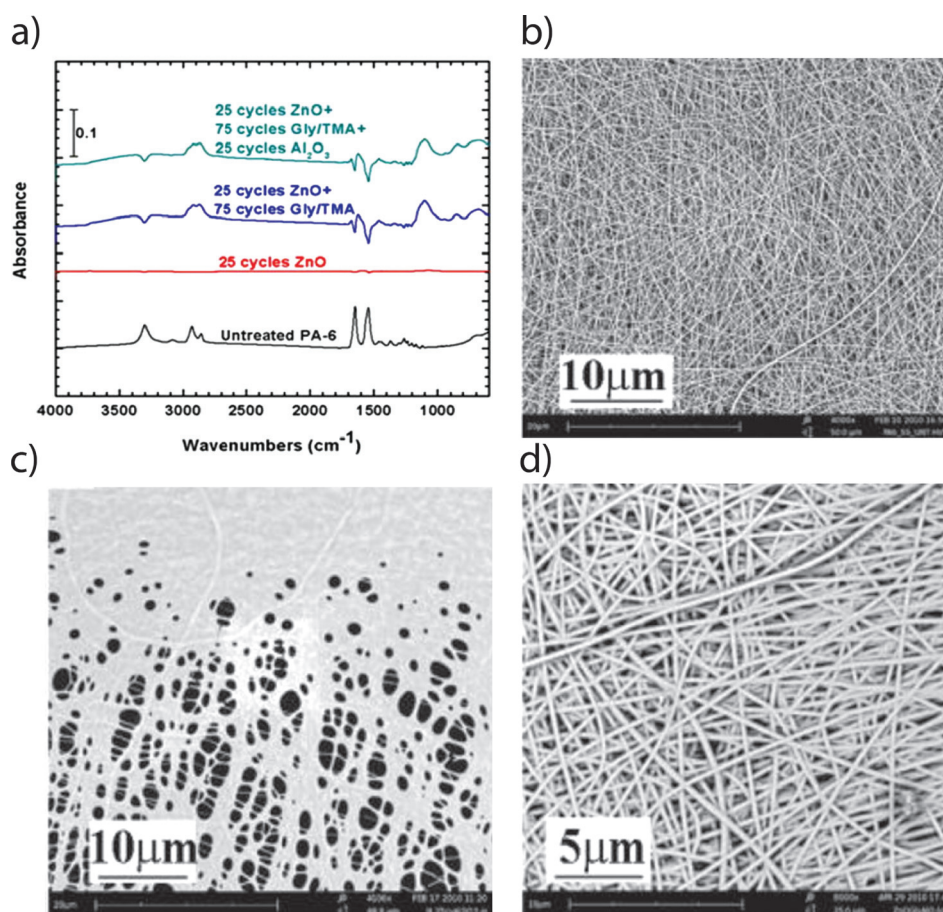


FIG. 7. (Color online) (a) FT-IR spectra of untreated nylon-6 nanofibers show almost no change after ZnO ALD coating, suggesting little chemical interaction of DEZ with the polyamide material. A subsequent glycidol/TMA MLD layer interacts with the carbonyl and amide modes, although no strong spectral difference is observed if an additional layer of Al₂O₃ follows the alucone, suggesting the MLD coating protects the nylon nanofibers from further degradation. (b) SEM images of the untreated nylon show that (c) degradation after TMA exposure is (d) minimized with the protective combination of ZnO and glycidol/TMA coating. Reprinted with permission from Oldham *et al.*, *J. Electrochem. Soc.* **158**, D549 (2011). Copyright 2011, The Electrochemical Society.

at 200 °C. The metal coating created a visible color change throughout the fiber mat, consistent with uniform coating. Cross-sectional TEM images also showed good conformality on the high surface area noncylindrical fibers used. After W ALD, the coated fibers retained the same flexibility of the uncoated samples and they readily returned to their starting shape after bending, compressing, or twisting. Quantitatively, the ALD-coated nylon mats maintained 90% of their conductivity after being flexed around a cylinder with a radius as small as 0.3 cm. Figure 9(a) shows a conductive flexible textile circuit using the ALD-coated nonwoven.¹⁰¹

Platinum coatings on fibers are of interest for high conductivity and catalysis. However, typical combustion-based Pt ALD often requires temperatures of 250–300 °C,^{148,149} which is too high for most textile materials. As shown in Fig. 9(b), Mackus *et al.* successfully demonstrated room-temperature Pt ALD onto cotton, paper, polyethylene terephthalate (PET), and polyethylene naphthalate fibers by dosing the Pt precursor, MeCpPtMe₃, followed by O₂ plasma treatment, and H₂ or H₂ plasma reduction.¹⁰⁴ They also found an Al₂O₃ precoat helped promote Pt nucleation. Analyses by SEM and EDX show good uniformity of the Pt coating, but the extent of coating into the fiber was not directly analyzed.¹⁰⁴

Gas sensors are a natural extension of ALD grown conductive coatings on fibers. The high surface area of fibrous materials, particularly natural materials such as cellulose, make attractive substrates for the deposition of conductive thin films that can provide sensing capabilities with improved sensitivity compared to planar substrates. For instance, analyte adsorption can alter the surface charge on a metal oxide leading to a measurable change in conductivity. Using electrospun polyacrylonitrile nanofibers as a starting substrate, Cho *et al.* used ZnO ALD coating followed by O₂ plasma etching and annealing to create ZnO gas sensing nanotubes. These structures showed a large decrease in electrical conductivity upon exposure to ethanol in air.¹⁵⁰ Kim *et al.*⁴⁴ also created a network of SnO₂ nanotubes that reversibly responded to various gases, including H₂, NH₃, NO₂, and CO. The highest gas responses were observed for ethanol—suggesting that these ALD template-synthesized nanotubes displayed good gas selectivity. Metal oxide nanofibers can also be directly formed by electrospinning from a metal-containing solution followed by calcination. Using this approach, Choi *et al.*³⁶ modified electrospun SnO₂ nanofibers using ZnO ALD, which showed good sensitivity for the detection of NO₂.

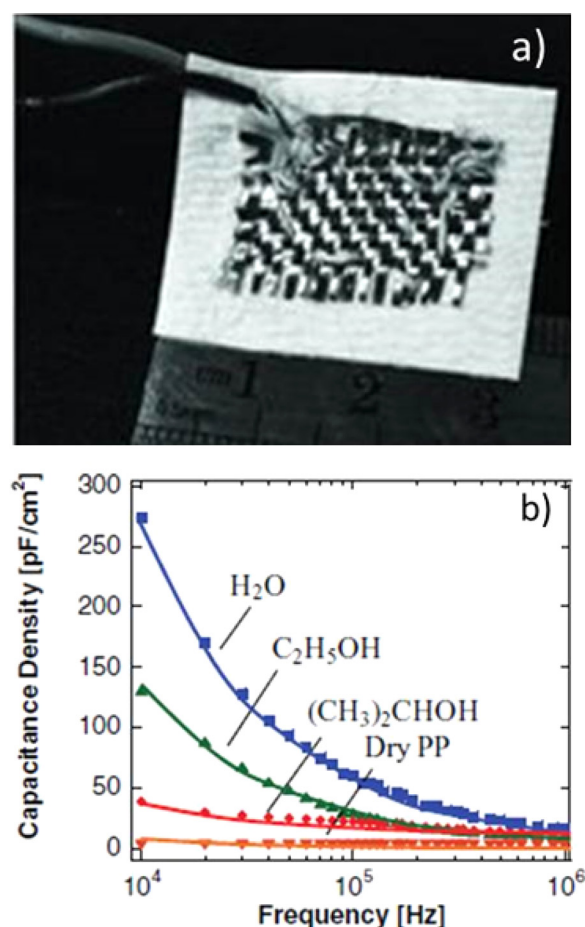


FIG. 8. (Color online) (a) Metal-insulator-metal fiber capacitor made from W coated fiber electrodes with a nonwoven polypropylene insulating layer. (b) Exposing the capacitor to various solvents created a capacitance signal that scaled with the solvent polarizability. Reprinted with permission from Jur *et al.*, *Adv. Funct. Mater.* **21**, 1993 (2011). Copyright 2011, John Wiley and Sons.

Other high surface metal oxide nanostructures can be made using metal vapor reaction into polymer fibers. Infusing various polyester fibers with TMA followed by calcination in air produced porous metal oxide monoliths that mimicked the shape of the starting fibers. The surface area and pore size distribution was tuned by changing the number dose cycles and the calcination conditions.¹²⁷ In another study, Heikkilä *et al.*¹⁵¹ found that dissolving polyvinyl alcohol nanofibers after ALD Al_2O_3 led to high surface area “nanocorals.” This unusual morphology was ascribed to migration of the polymer through cracks in the Al_2O_3 , transforming the ALD layer into a high surface area material, which could be useful for gas sensing applications.

In addition to gas sensors, conductive ALD coatings can also be used for fabric-integrated touch sensors.¹²⁴ Using conductive ALD Al-doped ZnO on spun-bonded nylon-6 nonwovens, Sweet *et al.* demonstrated an electronic fabric touch-sensor using a porous spacer between two conducting fabric layers. The device showed a large on/off ratio of up to 10^6 , and the signal response correlated well with the net applied force (Fig. 10). Such devices would have advantages in smart textile applications for the breathability, flexibility,

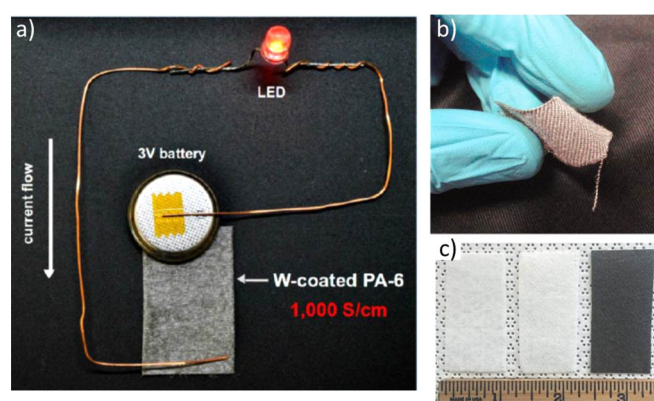


FIG. 9. (Color online) (a) W-coated nylon-6 fabric display excellent conductivity (1000 S/cm) in an active circuit. Reprinted with permission from Kalanyan *et al.*, *ACS Appl. Mater. Interfaces* **5**, 5253 (2013). Copyright 2013, American Chemical Society. (b) Pt Plasma ALD on woven cotton by and (c) ozone-based Pt ALD on nonwoven nylon. Reprinted with permission from Mackus *et al.*, *Chem. Mater.* **25**, 1769 (2013). Copyright 2013, American Chemical Society and Mundy *et al.* (Ref. 126), respectively.

and sensitivity needed for personalized diagnostics and monitoring.¹²⁴

Feature patterning is critical for electronic integration. Most wiring in electronic textiles uses metal lines that are sprayed, glued, stamped, printed, or otherwise secured onto one side of the support fabric, or full circuits that are secured to fabrics by heat-sealing the stitching. This works well for topically applied circuits, but true electronic textiles will build circuits within the textile, and may require patterned materials to extend throughout the fiber mat. For ALD on planar surfaces, “shadow” masking to create patterns does not work well because ALD reactants penetrate and deposit film in the thin gap region between the mask and the substrate. On fibrous substrates, precursor penetration under physical masks will be even more pronounced. However, if the fiber substrate is mechanically compressed by the mask, then lateral precursor transport can be slowed, creating uniform patterned coatings in the surface-exposed regions of the substrate. To demonstrate this, Sweet *et al.*¹⁰⁶ used physical masks to locally compress nonwoven and woven fabrics and study the extent of precursor transport into the compressed region during ALD of aluminum-doped ZnO. The random fiber network in the nonwoven helped impede lateral reactant diffusion and create more abrupt features, whereas the compressed woven fabric retained pathways for rapid lateral transport, leading to more diffuse patterns. A downside to this method is that the mask must be robust enough to apply mechanical force, so the minimum feature size will be limited. As shown in Fig. 11, features on the millimeter scale were demonstrated.¹⁰⁶ Alternatively, Mousa and Parsons¹⁵² used direct delivery of the ALD reactants to a local region on a fabric surface to locally coat porous fiber materials.

B. Energy conversion and electrochemical energy storage

Energy generation and storage are possibly the biggest challenges to full implementation of low-weight electronic

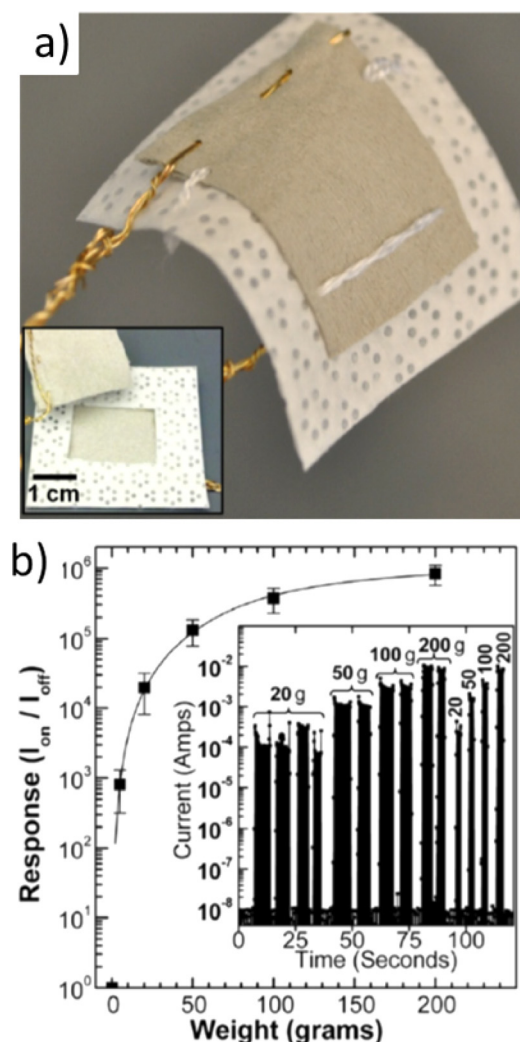


FIG. 10. (Color online) (a) Electronic fiber-based touch-sensor using nylon-6 fibers coated with ALD aluminum-doped ZnO separated by insulating porous nonwoven polypropylene. (b) The response current was highly reproducible and scaled directly with applied force over several orders of magnitude. Reprinted with permission from Sweet *et al.*, *J. Vac. Sci. Technol. Vac. Surf. Films* **33**, 01A117 (2015). Copyright 2015, AIP Publishing LLC.

textiles. Battery efficiencies are improved using high-surface area substrates to maximize the number of charge-transfer reactions. Permeation of electrolyte into a porous medium also improves battery performance by diminishing ion diffusion-path lengths. ALD has been proposed as an effective method of depositing conducting or electrochemically active materials for superior anodes, cathodes, and solid electrolytes.¹⁵³ For example, depositing ALD V_2O_5 on high surface area carbon can improve lithium and sodium ion batteries,^{39,154–156} and add pseudocapacitance to increase charge storage density in electrolytic capacitors.¹⁵⁷

In one example study, Chen *et al.* made paper substrates conductive with a carbon nanotube coating, and then added ALD V_2O_5 with an ozone-based process to create an electrolytic charge storage layer.^{82,156} The resulting structure gave a specific capacity of 321 mAh/g, which reflects the hierarchical pore structure in the paper. The macro- and mesopores, interfiber macropores, and the intrafiber mesopores

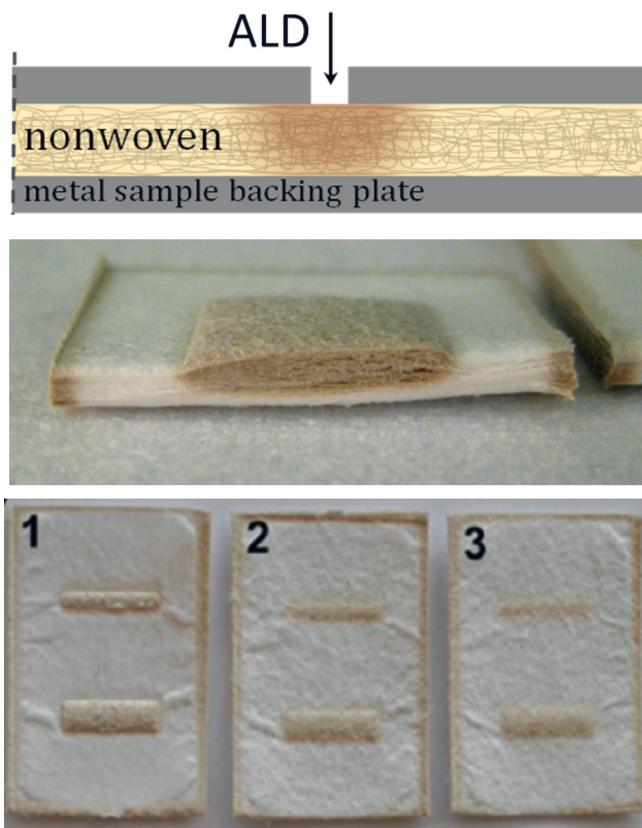


FIG. 11. (Color online) Patterned ALD ZnO on nonwoven nylon-6 using a compressive metal template. Brown features appear where ZnO has been deposited through the openings in the mask. The middle image shows a side-view of a nonwoven mat stack after ALD, and the bottom image shows the first, second and third nonwoven mat coated in a similar stack. The dose times during the ALD process allowed ZnO to coat through three stacked nonwoven mats. The patterns are ~ 2 mm wide. Reprinted with permission from Sweet *et al.*, *ACS Appl. Mater. Interfaces* **6**, 9280 (2014). Copyright 2014, American Chemical Society.

promote fast lithium ion diffusion between the electrolyte and the V_2O_5 . Macropores help prevent lithium ion depletion at the surface of the cathode material, which increases the charge/discharge rate.

C. Catalysis and photocatalysis

Catalytic and photocatalytic fibers and textiles is a highly active research area using ALD and other approaches.^{25,28,30,40,43,126,144,158–161} Cellulose is a particularly attractive substrate for catalysis and photocatalysis^{26,29,43,160} because of its high density of hydroxyl surface-functional groups, chemical and physical robustness, and its ability to withstand relatively high temperatures. Kemell *et al.*⁴³ showed that crystalline anatase TiO_2 could be formed on cellulose at 250 °C without significant degradation by using titanium ethoxide and water. The films were able to photocatalyze the deposition of silver from $Ag(NO_3)_3$ and the fiber/ TiO_2 network retained its flexibility and overall structure after coating. TiO_2 coated fibers were also shown to successfully photodegrade methylene blue dye in aqueous solution. Doping the TiO_2 ALD process with iridium improved photocatalytic behavior due to improved charge separation [Figs. 12(a) and 12(b)].¹⁶⁰

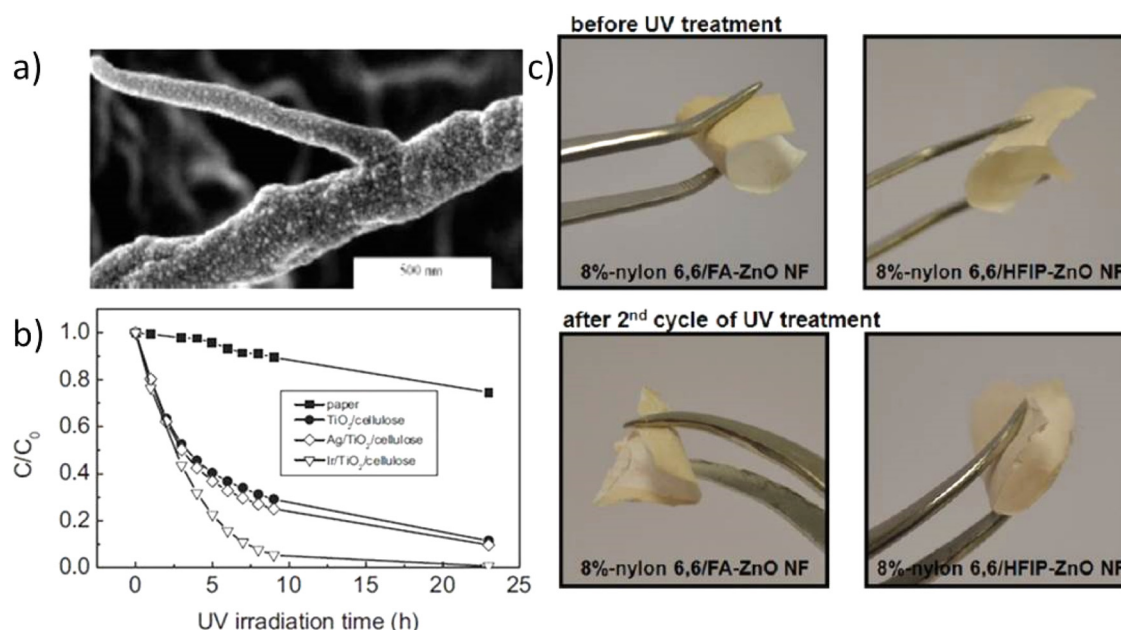


FIG. 12. (Color online) (a) SEM image of ALD TiO₂ on cellulose fibers shows a rough, high-surface area photocatalytic material. (b) Iridium doped TiO₂ on cellulose fibers increases the photocatalytic degradation of methylene blue. Reprinted with permission from Kemell *et al.*, Chem. Vap. Deposition **12**, 419 (2006). Copyright 2006, John Wiley and Sons. (c) Photocatalytic nylon 6, 6-ZnO core-shell nanofibers before UV treatment and after two cycles of UV treatment, showing flexibility is maintained. Reprinted with permission from Kayaci *et al.*, ACS Appl. Mater. Interfaces **4**, 6185 (2012). Copyright 2012, American Chemical Society.

Kayaci *et al.*⁴⁰ demonstrated photocatalytically active ZnO by ALD at 200 °C onto electrospun nylon 6,6. As shown in Fig. 12(c), the nanofiber/oxide composites remained flexible after use for photocatalysis. In other work,¹⁴⁴ the photocatalytic crystalline ZnO nanorods were formed on a polybutylene terephthalate nonwoven mat via hydrothermal processing, where ALD ZnO was used to create a nucleation layer to promote hydrothermal crystal growth. The polybutylene terephthalate mat alone did not permit crystal growth, in part, because its hydrophobic surface did not allow solution to penetrate between the fibers. However, adding an ALD layer made the fiber hydrophilic (as described above), helping to create a more uniform and dense layer of crystals growing normal to the fiber surface. Such catalytic and photocatalytic fabrics could be used in the future in defense applications, such as military-grade uniforms, to degrade toxic agents or chemical threats as they are encountered in potentially hostile environments, or alternatively these same materials could be used in filtration applications, such as for water-purification purposes.

A range of other studies have used ALD Pt on carbon nanotubes for proton exchange membranes, catalytic and electrocatalytic applications.^{162,163} Platinum ALD using trimethyl (methylcyclopentadienyl) platinum (IV) and O₂ at 270–300 °C generally grows via Volmer-Webber island growth, resulting in relatively high surface area Pt nuclei before complete film coalescence. At ~200 °C or less, Pt ALD can proceed on nylon or other fibers using ozone or plasma excited oxygen.^{104,126}

D. Chemically absorptive and reactive textiles

Metal-organic-frameworks (MOFs) are a novel class of materials that utilize metal-organic ligand interactions to

generate micro- and nanoscale sized crystals with tunable pore sizes that result in record high porosity values. Because of the high porosity and surface area, MOFs have been proposed for filtration, separation, sensing, and gas storage applications.¹⁶⁴ However, these uses have been slow to develop, in part, due to the limitations of MOF processing in the powder phase.

Zhao *et al.* show that ALD coatings on polymer fibers can help promote nucleation of covalently bound HKUST-1 MOF crystals, producing a multifunctional MOF/fiber composite fabric by hydrothermal and layer-by-layer synthesis.^{117,146} Using hydrothermal synthesis at 120 °C, polypropylene nonwoven microfibers coated with ALD Al₂O₃ showed MOF mass uptake >50% larger than that on untreated fibers (Fig. 13).¹⁴⁶ Moreover, the ALD coating promoted heterogeneous MOF nucleation, whereas uncoated fibers led to more homogeneous nucleation, leading to nonattached MOF crystals between the fibers. The MOF/fiber composites showed an average surface area close to 700 m²/g, corresponding to >900 m²/g of MOF, consistent with that measured on similarly prepared crystal powders. The resulting mats showed NH₃ dynamic loading >5 mol/kg_{MOF+fiber}, exceeding previous reports for absorptive MOFs on fibers. Other fiber materials including PET and cellulose cotton also showed MOF uptake, although the MOF growth on cotton was less than on the other materials. The authors hypothesized that the Al₂O₃ on cotton promoted MOF nucleation, with poorer adhesion on the smooth cotton fiber surface. ALD ZnO and TiO₂ also showed good results with a generally larger HKUST-1 mass uptake on the ZnO relative to Al₂O₃ and TiO₂. Beyond HKUST-1, other MOF materials including UiO-66 and MOF-74 also showed good growth on ALD-coated fibers.¹⁴⁶ More detailed

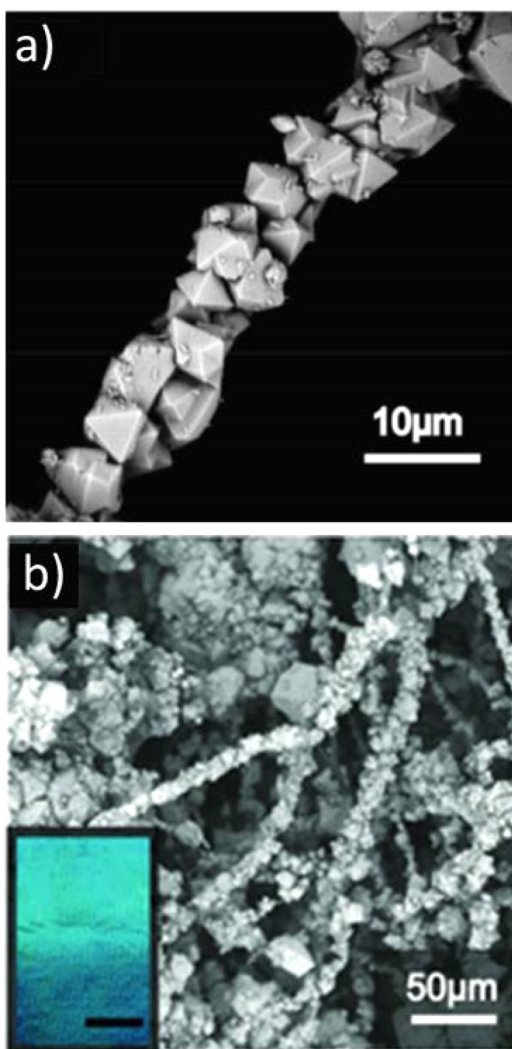


FIG. 13. (Color online) SEM images of HKUST-1 MOF grown on polymer fibers pretreated with 200 cycles of Al_2O_3 ALD. (a) An individual polypropylene fiber with HKUST-1 MOFs and (b) the corresponding fiber mat imaged using an SEM and (inset) optical microscope. The scale bar in the inset is 1 cm. Reprinted with permission from Zhao *et al.*, *Adv. Mater. Interfaces* **1**, 1400040 (2014). Copyright 2014, John Wiley and Sons.

analysis of MOF nucleation and reaction mechanisms revealed that the $\text{Cu}(\text{NO}_3)_2$ MOF precursor could react with ZnO to form an intermediate layered hydroxyl-metal salt that allowed rapid anion exchange to form the HKUST-1 crystal.

As a multifunctional textile, MOF loaded fabrics could be essential as an added line of defense against chemical warfare agents by utilizing the ultrahigh porosity to absorb such harmful compounds from the atmosphere. Alternatively, ALD-MOF modified fibers and nonwoven mats could be used as the active material in filtration media to similarly remove unwanted materials. The ability of ALD to improve the adherence of MOFs to fibrous materials helps extend the range to which these exciting new class of chemical compounds can be applied.

VI. OUTLOOK AND CHALLENGES FOR ALD ON TEXTILES

Fiber and textile materials found throughout daily life are emerging as a versatile platform to introduce electronic or

other nonaesthetic multifunctional capabilities that are easy to use and blend in with current clothing trends. As shown above, ALD is a promising technology to modify, improve, or otherwise add new performance capability to textiles. The promise of ALD for textiles stems from how the coating deposits on fibers and fabrics. Using controlled self-limiting ALD nucleation and growth conditions, textile materials are coated with a uniform thin shell of inorganic or organic material that leaves the weight, feel, and look generally unchanged. Other ALD-based processes may also be important to modify the near-surface or bulk materials of fibrous polymers.

The most demanding challenge for ALD in textiles is that common batch processing is too slow and costly for most applications. Therefore, early adoption of ALD will likely occur in high-value products such as specialty medical materials, high-performance aerospace composites, or advanced military applications where current fiber treatment processes are not sufficiently uniform or consistent to meet the product needs.

To fully realize the potential for ALD on textiles, new high-throughput processing methods are needed. The speed and cost required for ALD on textiles will depend on the industry and product target. Textile finishes for woven garments often run at line speeds of 10–20 m/min, which is close to many current spatial ALD deposition speeds. On the other hand, nonwovens are becoming a larger part of the textile industry because of low cost and high throughput, often exceeding 100 m/min. Continued advances in spatial and roll-to-roll ALD,^{165–168} flow-through coating processes, and the extension of spatial ALD to new materials will be important. The combination of high-surface area, porosity, and tortuosity found in fiber-networks will require better understanding of reactant transport and reaction on fibers, kinetics of byproduct desorption and diffusion, and improved reactor designs for ambient pressure operation (Fig. 14). Textile manufacturing is one of the oldest product industries, and many barriers will impede new technology adoption. Integrating a process like ALD into manufacturing plants with uncontrolled humidity, dust, and other contaminants will be challenging.

After surmounting these issues, early applications for lower-cost ALD on fibers will likely augment or improve current textile surface-treatments for chemical functionalization. Surface finishes on nonreactive polypropylene often rely on corona discharge or plasmas to attach temporary

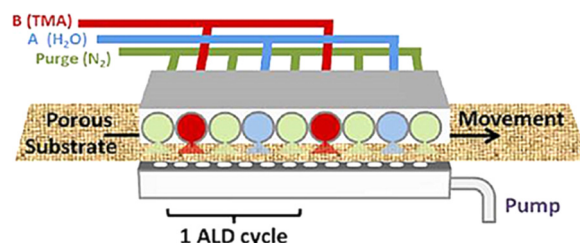


FIG. 14. (Color online) Schematic illustration of a spatial ALD reactor for roll-to-roll ALD on textiles. Continuous flow of TMA and water, separated by inert gas inlet ports allow reactants to flow through the porous substrate to the exhaust located under the fiber mat.

bonding sites to one side of the fabric. For current solution-based methods, surface modification by ALD could reduce chemical use and remove energy-intensive drying steps. ALD and MLD processes are chemically flexible, and all fibers and surfaces in contact with the precursors are coated with the desired chemistry, creating uniform sites for chemical grafting of amides, silanes, fluorinated groups, and other compounds. These can be finishers or primers for further reaction to improve adhesion, modify electrostatic interaction, surface wetting or appearance, or add biocidal or antimicrobial finishes.

Beyond preparing textiles for further treatment, ALD coatings themselves could make significant improvements in fiber and textile-based products. ALD layers can protect polymers from UV degradation, but cost and throughput will again be key challenges. Biocompatibility is an opportunity area, but more work is needed in ALD/biomaterial interactions. ALD materials could be particularly helpful in advanced filtration technology. A challenge in filtration is to improve capture or chemical degradation capacity while maintaining vapor or liquid throughput for a given pressure drop.¹⁶⁹ Nanoscale ALD coatings on polymer fiber filters can broaden performance in chemically harsh environments while maintaining good species transport performance. Coupling new functional catalysts, including MOF materials, onto fibers will also lead to important new filtration and chemical mitigation applications. Further benefits of ALD coatings for filtration include reduced fouling, reduction or stabilization of static charge, and integration of new materials for real-time chemical sensing capacity.

Further out, future advances in ALD metals, semiconductors, or other materials could enable textile-based switching, energy harvesting, energy storage, information display, chemical or physical sensing, and integrated communication. New low-temperature ALD processes, for example, for piezoelectric, thermoelectric,¹⁷⁰ photoactive perovskite materials,¹⁷¹ or electropositive metals¹⁷² could be important. MOFs and other porous materials could expand sensing, catalysis, and gas storage applications. Other new chemically active materials could create responsive textiles that change porosity or thermal insulation using input from the environment, or change the look or feel of a material in response to a user's needs.

Along with the need for new materials, more understanding is needed for most every ALD process regarding precursor interactions with natural and synthetic materials, including microscopic processes and mechanisms in vapor infiltration and polymer metalation. Additionally, understanding how ALD textiles function and wear in real-world environments, including machine washing and dry-cleaning,¹⁷³ remains a key question and potential challenge. Overcoming these hurdles will help advance many ideas for smart, electronic and multifunctional textiles that promise to change the way we interact with each other and with our environment.

ACKNOWLEDGMENT

The authors acknowledge support from the National Science Foundation, Grant No. 1344618.

- ¹R. F. Service, *Science* **301**, 909 (2003).
- ²B. S. Shim, W. Chen, C. Doty, C. Xu, and N. A. Kotov, *Nano Lett.* **8**, 4151 (2008).
- ³K. Cherenack, C. Zysset, T. Kinkeldei, N. Muenzenrieder, and G. Troester, *Adv. Mater.* **22**, 5178 (2010).
- ⁴K. Cherenack and L. van Pieterse, *J. Appl. Phys.* **112**, 091301 (2012).
- ⁵I. S. Bayer and A. Biswas, *Vac. Technol. Coat.* **14**, 14 (2013).
- ⁶M. Stoppa and A. Chiolerio, *Sensors* **14**, 11957 (2014).
- ⁷E. R. Post, M. Orth, P. R. Russo, and N. Gershenfeld, *IBM Syst. J.* **39**, 840 (2000).
- ⁸D.-H. Kim, Y.-S. Kim, J. Wu, Z. Liu, J. Song, H.-S. Kim, Y. Y. Huang, K.-C. Hwang, and J. A. Rogers, *Adv. Mater.* **21**, 3703 (2009).
- ⁹G. S. Ryu, S. H. Jeong, B. C. Park, B. Park, and C. K. Song, *Org. Electron.* **15**, 1672 (2014).
- ¹⁰H. F. Castro, E. Sowade, J. G. Rocha, P. Alpuim, A. V. Machado, R. R. Baumann, and S. Lanceros-Méndez, *Org. Electron.* **22**, 12 (2015).
- ¹¹R. Bhattacharya, M. M. de Kok, and J. Zhou, *Appl. Phys. Lett.* **95**, 223305 (2009).
- ¹²Y. Fu, X. Cai, H. Wu, Z. Lv, S. Hou, M. Peng, X. Yu, and D. Zou, *Adv. Mater.* **24**, 5713 (2012).
- ¹³T. Yamada, Y. Hayamizu, Y. Yamamoto, Y. Yomogida, A. Izadi-Najafabadi, D. N. Futaba, and K. Hata, *Nat. Nanotechnol.* **6**, 296 (2011).
- ¹⁴X. Cai, M. Peng, X. Yu, Y. Fu, and D. Zou, *J. Mater. Chem. C* **2**, 1184 (2014).
- ¹⁵M. Hamed, R. Forchheimer, and O. Inganäs, *Nat. Mater.* **6**, 357 (2007).
- ¹⁶W. Zeng, L. Shu, Q. Li, S. Chen, F. Wang, and X.-M. Tao, *Adv. Mater.* **26**, 5310 (2014).
- ¹⁷R. He, T. D. Day, M. Krishnamurthi, J. R. Sparks, P. J. A. Sazio, V. Gopalan, and J. V. Badding, *Adv. Mater.* **25**, 1461 (2013).
- ¹⁸D. Graham-Rowe, *Nat. Photonics* **1**, 6 (2007).
- ¹⁹A. Laforgue, *J. Mater. Chem.* **20**, 8233 (2010).
- ²⁰L. Liu, Y. Yu, C. Yan, K. Li, and Z. Zheng, *Nat. Commun.* **6**, 7260 (2015).
- ²¹D. Zou, D. Wang, Z. Chu, Z. Lv, and X. Fan, *Coord. Chem. Rev.* **254**, 1169 (2010).
- ²²M. Toivola, M. Ferenets, P. Lund, and A. Harlin, *Thin Solid Films* **517**, 2799 (2009).
- ²³J. Lee *et al.*, *Adv. Mater.* **27**, 2433 (2015).
- ²⁴Y. Huang, *et al.*, *ACS Nano* **9**, 4766 (2015).
- ²⁵B. Yang, C. Zhao, M. Xiao, F. Wang, C. Li, J. Wang, and J. C. Yu, *Small* **9**, 1003 (2013).
- ²⁶L. Karimi, M. E. Yazdanzhenas, R. Khajavi, A. Rashidi, and M. Mirjalili, *Cellulose* **21**, 3813 (2014).
- ²⁷J. Kiwi and C. Pulgarin, *Catal. Today* **151**, 2 (2010).
- ²⁸J.-W. Lee, T. Mayer-Gall, K. Opwis, C. E. Song, J. S. Gutmann, and B. List, *Science* **341**, 1225 (2013).
- ²⁹G. Goncalves, P. A. A. P. Marques, R. J. B. Pinto, T. Trindade, and C. P. Neto, *Compos. Sci. Technol.* **69**, 1051 (2009).
- ³⁰J. Molina, F. Fernandes, J. Fernández, M. Pastor, A. Correia, A. P. Souto, J. O. Carneiro, V. Teixeira, and F. Cases, *Mater. Sci. Eng. B* **199**, 62 (2015).
- ³¹R. Dastjerdi and M. Montazer, *Colloids Surf. B Biointerfaces* **79**, 5 (2010).
- ³²S. D. McCullen, D. R. Stevens, W. A. Roberts, L. I. Clarke, S. H. Bernacki, R. E. Gorga, and E. G. Lobo, *Int. J. Nanomed.* **2**, 253 (2007).
- ³³Z. Ma, M. Kotaki, R. Inai, and S. Ramakrishna, *Tissue Eng.* **11**, 101 (2005).
- ³⁴Y.-P. Jiao and F.-Z. Cui, *Biomed. Mater.* **2**, R24 (2007).
- ³⁵J. S. Jur, J. C. Spagnola, K. Lee, B. Gong, Q. Peng, and G. N. Parsons, *Langmuir* **26**, 8239 (2010).
- ³⁶S.-W. Choi, J. Y. Park, and S. S. Kim, *Nanotechnology* **20**, 465603 (2009).
- ³⁷N. A. Burns, P. Williams, A. H. Brozena, D. Sen, S. Atanasov, G. N. Parsons, and S. A. Khan, *Adv. Mater. Interfaces* **2**, 1500229 (2015).
- ³⁸Z. Gui, H. Zhu, E. Gillette, X. Han, G. W. Rubloff, L. Hu, and S. B. Lee, *ACS Nano* **7**, 6037 (2013).
- ³⁹X. Chen, H. Zhu, Y.-C. Chen, Y. Shang, A. Cao, L. Hu, and G. W. Rubloff, *ACS Nano* **6**, 7948 (2012).
- ⁴⁰F. Kayaci, C. Ozgit-Akgun, I. Donmez, N. Biyikli, and T. Uyar, *ACS Appl. Mater. Interfaces* **4**, 6185 (2012).
- ⁴¹G. N. Parsons *et al.*, *Coord. Chem. Rev.* **257**, 3323 (2013).
- ⁴²C. A. Wilson, R. K. Grubbs, and S. M. George, *Chem. Mater.* **17**, 5625 (2005).

- ⁴³M. Kemell, V. Pore, M. Ritala, M. Leskelä, and M. Lindén, *J. Am. Chem. Soc.* **127**, 14178 (2005).
- ⁴⁴W.-S. Kim, B.-S. Lee, D.-H. Kim, H.-C. Kim, W.-R. Yu, and S.-H. Hong, *Nanotechnology* **21**, 245605 (2010).
- ⁴⁵J. Musschoot, J. Dendooven, D. Deduytsche, J. Haemers, G. Buyle, and C. Detavernier, *Surf. Coat. Technol.* **206**, 4511 (2012).
- ⁴⁶A. K. Roy *et al.*, *Anal. Bioanal. Chem.* **396**, 1913 (2010).
- ⁴⁷L. Seung-Mo, E. Pippel, U. Gösele, C. Dresbach, Y. Qin, C. V. Chandran, T. Brauniger, G. Hause, and M. Knez, *Science* **324**, 488 (2009).
- ⁴⁸R. Nayak, R. Padhye, I. L. Kyratzis, Y. B. Truong, and L. Arnold, *Text. Res. J.* **82**, 129 (2012).
- ⁴⁹L. Huang, R. A. McMillan, R. P. Apkarian, B. Pourdeyimi, V. P. Conticello, and E. L. Chaikof, *Macromolecules* **33**, 2989 (2000).
- ⁵⁰N. Fedorova and B. Pourdeyimi, *J. Appl. Polym. Sci.* **104**, 3434 (2007).
- ⁵¹J. H. Wendorff, S. Agarwal, and A. Greiner, *Electrospinning: Materials, Processing, and Applications* (Wiley, Weinheim, Germany, 2012).
- ⁵²C. Burger, B. S. Hsiao, and B. Chu, *Annu. Rev. Mater. Res.* **36**, 333 (2006).
- ⁵³P. Bajaj, *J. Appl. Polym. Sci.* **83**, 631 (2002).
- ⁵⁴S. Gowri, L. Almeida, T. Amorim, N. Carneiro, A. P. Souto, and M. F. Esteves, *Text. Res. J.* **80**, 1290 (2010).
- ⁵⁵B. Mahltig, H. Haufe, and H. Böttcher, *J. Mater. Chem.* **15**, 4385 (2005).
- ⁵⁶C. Drew, X. Liu, D. Ziegler, R. Wang, F. F. Bruno, J. Whitten, L. A. Samuelson, and J. Kumar, *Nano Lett.* **3**, 143 (2003).
- ⁵⁷J. Liu, Q. Wang, and X. R. Fan, *J. Sol-Gel Sci. Technol.* **62**, 338 (2012).
- ⁵⁸L. Hu, M. Pasta, F. La Mantia, L. Cui, S. Jeong, H. D. Deshazer, J. W. Choi, S. M. Han, and Y. Cui, *Nano Lett.* **10**, 708 (2010).
- ⁵⁹Y. Sun and G. Sun, *J. Appl. Polym. Sci.* **84**, 1592 (2002).
- ⁶⁰K. C. Krogman, N. S. Zacharia, S. Schroeder, and P. T. Hammond, *Langmuir* **23**, 3137 (2007).
- ⁶¹K. C. Krogman, J. L. Lowery, N. S. Zacharia, G. C. Rutledge, and P. T. Hammond, *Nat. Mater.* **8**, 512 (2009).
- ⁶²S. S. Chhatre, A. Tuteja, W. Choi, A. Revaux, D. Smith, J. M. Mabry, G. H. McKinley, and R. E. Cohen, *Langmuir* **25**, 13625 (2009).
- ⁶³W. Choi, A. Tuteja, S. Chhatre, J. M. Mabry, R. E. Cohen, and G. H. McKinley, *Adv. Mater.* **21**, 2190 (2009).
- ⁶⁴T. Wakida, S. Tokino, Shouhua Niu, H. Kawamura, Y. Sato, M. Lee, H. Uchiyama, and H. Inagaki, *Text. Res. J.* **63**, 433 (1993).
- ⁶⁵R. Morent, N. De Geyter, J. Verschuren, K. De Clerck, P. Kiekens, and C. Leys, *Surf. Coat. Technol.* **202**, 3427 (2008).
- ⁶⁶H. U. Poll, U. Schladitz, and S. Schreiter, *Surf. Coat. Technol.* **142**, 489 (2001).
- ⁶⁷M. Lee, M. S. Lee, T. Wakida, T. Tokuyama, G. Inoue, S. Ishida, T. Itazu, and Y. Miyaji, *J. Appl. Polym. Sci.* **100**, 1344 (2006).
- ⁶⁸M. Ma, Y. Mao, M. Gupta, K. K. Gleason, and G. C. Rutledge, *Macromolecules* **38**, 9742 (2005).
- ⁶⁹H. Szymanowski, A. Sobczyk, M. Gazicki-Lipman, W. Jakubowski, and L. Klimek, *Surf. Coat. Technol.* **200**, 1036 (2005).
- ⁷⁰J. Scholz, G. Nocke, F. Hollstein, and A. Weissbach, *Surf. Coat. Technol.* **192**, 252 (2005).
- ⁷¹R. P. Seiber and H. L. Needles, *J. Appl. Polym. Sci.* **19**, 2187 (1975).
- ⁷²A. A. Armstrong and H. A. Rutherford, *Text. Res. J.* **33**, 264 (1963).
- ⁷³W. E. Tenhaeff and K. K. Gleason, *Adv. Funct. Mater.* **18**, 979 (2008).
- ⁷⁴S. M. George, *Chem. Rev.* **110**, 111 (2010).
- ⁷⁵M. Knez, K. Nielsch, and L. Niinistö, *Adv. Mater.* **19**, 3425 (2007).
- ⁷⁶G. N. Parsons, S. M. George, and M. Knez, *MRS Bull.* **36**, 865 (2011).
- ⁷⁷M. Ritala, M. Kemell, M. Lautala, A. Niskanen, M. Leskelä, and S. Lindfors, *Chem. Vap. Deposition* **12**, 655 (2006).
- ⁷⁸G. N. Parsons, *Atomic Layer Deposition of Nanostructured Materials*, edited by N. Pinna and M. Knez (Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim, Germany, 2011), pp. 271–300.
- ⁷⁹J. S. Jur, W. J. Sweet, C. J. Oldham, and G. N. Parsons, *Adv. Funct. Mater.* **21**, 1993 (2011).
- ⁸⁰C. A. Hanson, C. J. Oldham, and G. N. Parsons, *J. Vac. Sci. Technol. Vac. Surf. Films* **30**, 01A117 (2012).
- ⁸¹K. Lahtinen, P. Johansson, T. Kääriäinen, and D. C. Cameron, *Polym. Eng. Sci.* **52**, 1985 (2012).
- ⁸²X. Chen, H. Zhu, C. Liu, Y.-C. Chen, N. Weadock, G. Rubloff, and L. Hu, *J. Mater. Chem. A* **1**, 8201 (2013).
- ⁸³J. C. Spagnola, B. Gong, S. A. Arvidson, J. S. Jur, S. A. Khan, and G. N. Parsons, *J. Mater. Chem.* **20**, 4213 (2010).
- ⁸⁴W. J. Sweet, J. S. Jur, and G. N. Parsons, *J. Appl. Phys.* **113**, 194303 (2013).
- ⁸⁵B. Gong, J. C. Spagnola, S. A. Arvidson, S. A. Khan, and G. N. Parsons, *Polymer* **53**, 4631 (2012).
- ⁸⁶F. Bosco, R. A. Carletto, J. Alongi, L. Marmo, A. Di Blasio, and G. Malucelli, *Carbohydr. Polym.* **94**, 372 (2013).
- ⁸⁷J. W. Cho and D. R. Paul, *Polymer* **42**, 1083 (2001).
- ⁸⁸M. A. L. Manchado, L. Valentini, J. Biagiotti, and J. M. Kenny, *Carbon* **43**, 1499 (2005).
- ⁸⁹J. M. G. Cowie, *Eur. Polym. J.* **9**, 1041 (1973).
- ⁹⁰J. Rasburn, P. J. Hine, I. M. Ward, R. H. Olley, D. C. Bassett, and M. A. Kabeel, *J. Mater. Sci.* **30**, 615 (1995).
- ⁹¹M. Gilbert and F. J. Hybart, *Polymer* **13**, 327 (1972).
- ⁹²P. J. Hine, A. Astruc, and I. M. Ward, *J. Appl. Polym. Sci.* **93**, 796 (2004).
- ⁹³G. O. Shonaike, *Eur. Polym. J.* **28**, 777 (1992).
- ⁹⁴X.-G. Li and M.-R. Huang, *J. Appl. Polym. Sci.* **71**, 565 (1999).
- ⁹⁵B. J. Holland and J. N. Hay, *Polymer* **42**, 6775 (2001).
- ⁹⁶I. Sakurada, *Polyvinyl Alcohol Fibers* (Marcel Dekker, Inc., New York, 1985).
- ⁹⁷J. N. Hay, *J. Polym. Sci.* **6**, 2127 (1968).
- ⁹⁸W. H. Howard, *J. Appl. Polym. Sci.* **5**, 303 (1961).
- ⁹⁹L. He, Q. Xu, C. Hua, and R. Song, *Polym. Compos.* **31**, 921 (2009).
- ¹⁰⁰R. Gregorio, Jr and E. M. Ueno, *J. Mater. Sci.* **34**, 4489 (1999).
- ¹⁰¹B. Kalanyan, C. J. Oldham, W. J. Sweet, and G. N. Parsons, *ACS Appl. Mater. Interfaces* **5**, 5253 (2013).
- ¹⁰²A. Niskanen, K. Arstila, M. Ritala, and M. Leskelä, *J. Electrochem. Soc.* **152**, F90 (2005).
- ¹⁰³A. Niskanen, K. Arstila, M. Leskelä, and M. Ritala, *Chem. Vap. Deposition* **13**, 152 (2007).
- ¹⁰⁴A. J. M. Mackus, D. Garcia-Alonso, H. C. M. Knoops, A. A. Bol, and W. M. M. Kessels, *Chem. Mater.* **25**, 1769 (2013).
- ¹⁰⁵G. K. Hyde *et al.*, *Langmuir* **26**, 2550 (2010).
- ¹⁰⁶W. J. Sweet, C. J. Oldham, and G. N. Parsons, *ACS Appl. Mater. Interfaces* **6**, 9280 (2014).
- ¹⁰⁷Q. Peng, X.-Y. Sun, J. C. Spagnola, G. K. Hyde, R. J. Spontak, and G. N. Parsons, *Nano Lett.* **7**, 719 (2007).
- ¹⁰⁸G. K. Hyde, K. J. Park, S. M. Stewart, J. P. Hinestroza, and G. N. Parsons, *Langmuir* **23**, 9844 (2007).
- ¹⁰⁹C. D. McClure, C. J. Oldham, H. J. Walls, and G. N. Parsons, *J. Vac. Sci. Technol. Vac. Surf. Films* **31**, 061506 (2013).
- ¹¹⁰B. Gong and G. N. Parsons, *J. Mater. Chem.* **22**, 15672 (2012).
- ¹¹¹C. K. Devine, C. J. Oldham, J. S. Jur, B. Gong, and G. N. Parsons, *Langmuir* **27**, 14497 (2011).
- ¹¹²J. W. Elam, D. Routkevitch, P. P. Mardilovich, and S. M. George, *Chem. Mater.* **15**, 3507 (2003).
- ¹¹³S. O. Kucheyev, J. Biener, T. F. Baumann, Y. M. Wang, A. V. Hamza, Z. Li, D. K. Lee, and R. G. Gordon, *Langmuir* **24**, 943 (2008).
- ¹¹⁴N. P. Kobayashi, C. L. Donley, S.-Y. Wang, and R. S. Williams, *J. Cryst. Growth* **299**, 218 (2007).
- ¹¹⁵M. Ghanashyam Krishna, M. Vinjanampati, and D. Dhar Purkayastha, *Eur. Phys. J. Appl. Phys.* **62**, 30001 (2013).
- ¹¹⁶C. J. Oldham, B. Gong, J. C. Spagnola, J. S. Jur, K. J. Senecal, T. A. Godfrey, and G. N. Parsons, *J. Electrochem. Soc.* **158**, D549 (2011).
- ¹¹⁷K. Lee, J. S. Jur, D. H. Kim, and G. N. Parsons, *J. Vac. Sci. Technol. A* **30**, 01A163 (2012).
- ¹¹⁸A. B. D. Cassie and S. Baxter, *Trans. Faraday Soc.* **40**, 0546 (1944).
- ¹¹⁹R. N. Wenzel, *Ind. Eng. Chem.* **28**, 988 (1936).
- ¹²⁰G. K. Hyde, S. D. McCullen, S. Jeon, S. M. Stewart, H. Jeon, E. G. Loba, and G. N. Parsons, *Biomed. Mater.* **4**, 025001 (2009).
- ¹²¹*Wearable Electronics and Photonics*, edited by X. M. Tao (CRC, Boca Raton, FL, 2000).
- ¹²²J. J. Ge, H. Hou, Q. Li, M. J. Graham, A. Greiner, D. H. Reneker, F. W. Harris, and S. Z. D. Cheng, *J. Am. Chem. Soc.* **126**, 15754 (2004).
- ¹²³C. Xiang, W. Lu, Y. Zhu, Z. Sun, Z. Yan, C.-C. Hwang, and J. M. Tour, *ACS Appl. Mater. Interfaces* **4**, 131 (2012).
- ¹²⁴W. J. Sweet, C. J. Oldham, and G. N. Parsons, *J. Vac. Sci. Technol. Vac. Surf. Films* **33**, 01A117 (2015).
- ¹²⁵W. J. Sweet and G. N. Parsons, *Langmuir* **31**, 7274 (2015).
- ¹²⁶J. Z. Mundy, A. Shafiearhood, F. Li, S. A. Khan, and G. N. Parsons, *J. Vac. Sci. Technol. A* **34**, 01A152 (2015).
- ¹²⁷B. Gong, Q. Peng, J. S. Jur, C. K. Devine, K. Lee, and G. N. Parsons, *Chem. Mater.* **23**, 3476 (2011).

- ¹²⁸K. E. Gregorczyk, D. F. Pickup, M. G. Sanz, I. A. Irakulis, C. Rogero, and M. Knez, *Chem. Mater.* **27**, 181 (2015).
- ¹²⁹H. I. Akyildiz, R. P. Padbury, G. N. Parsons, and J. S. Jur, *Langmuir* **28**, 15697 (2012).
- ¹³⁰H. I. Akyildiz, M. B. M. Mousa, and J. S. Jur, *J. Appl. Phys.* **117**, 045301 (2015).
- ¹³¹D. S. Finch, T. Oreskovic, K. Ramadurai, C. F. Herrmann, S. M. George, and R. L. Mahajan, *J. Biomed. Mater. Res. A* **87**, 100 (2008).
- ¹³²M. Putkonen, T. Sajavaara, P. Rahkila, L. Xu, S. Cheng, L. Niinistö, and H. J. Whitlow, *Thin Solid Films* **517**, 5819 (2009).
- ¹³³A. J. Taylor, C. D. McClure, K. A. Shipkowski, E. A. Thompson, S. Hussain, S. Garantziotis, G. N. Parsons, and J. C. Bonner, *PLoS One* **9**, e106870 (2014).
- ¹³⁴S. E. Atanasov *et al.*, *J. Mater. Chem. A* **2**, 17371 (2014).
- ¹³⁵M. D. Groner, S. M. George, R. S. McLean, and P. F. Garcia, *Appl. Phys. Lett.* **88**, 051907 (2006).
- ¹³⁶S. M. George, B. Yoon, and A. A. Dameron, *Acc. Chem. Res.* **42**, 498 (2009).
- ¹³⁷P. Sundberg and M. Karppinen, *Beilstein J. Nanotechnol.* **5**, 1104 (2014).
- ¹³⁸M. Vaha-Nissi, P. Sundberg, E. Kauppi, T. Hirvikorpi, J. Sievanen, A. Sood, M. Karppinen, and A. Harlin, *Thin Solid Films* **520**, 6780 (2012).
- ¹³⁹S.-H. Jen, B. H. Lee, S. M. George, R. S. McLean, and P. F. Garcia, *Appl. Phys. Lett.* **101**, 234103 (2012).
- ¹⁴⁰Z. Ma, M. Kotaki, T. Yong, W. He, and S. Ramakrishna, *Biomaterials* **26**, 2527 (2005).
- ¹⁴¹B.-Y. Lee, K. Behler, M. E. Kurtoglu, M. A. Wynosky-Dolfi, R. F. Rest, and Y. Gogotsi, *J. Nanopart. Res.* **12**, 2511 (2010).
- ¹⁴²Q. Peng, B. Gong, and G. N. Parsons, *Nanotechnology* **22**, 155601 (2011).
- ¹⁴³J. Huang, I. Ichinose, and T. Kunitake, *Angew. Chem. Int. Ed.* **45**, 2883 (2006).
- ¹⁴⁴B. Gong, Q. Peng, J.-S. Na, and G. N. Parsons, *Appl. Catal. Gen.* **407**, 211 (2011).
- ¹⁴⁵F. Kayaci, S. Vempati, C. Ozgit-Akgun, N. Biyikli, and T. Uyar, *Appl. Catal. B-Environ.* **156**, 173 (2014).
- ¹⁴⁶J. Zhao *et al.*, *Adv. Mater. Interfaces* **1**, 1400040 (2014).
- ¹⁴⁷J. Zhao *et al.*, *J. Mater. Chem. A* **3**, 1458 (2015).
- ¹⁴⁸T. Aaltonen, M. Ritala, T. Sajavaara, J. Keinonen, and M. Leskelä, *Chem. Mater.* **15**, 1924 (2003).
- ¹⁴⁹T. Aaltonen, M. Ritala, Y.-L. Tung, Y. Chi, K. Arstila, K. Meinander, and M. Leskelä, *J. Mater. Res.* **19**, 3353 (2004).
- ¹⁵⁰S. Cho, D.-H. Kim, B.-S. Lee, J. Jung, W.-R. Yu, S.-H. Hong, and S. Lee, *Sens. Actuators B Chem.* **162**, 300 (2012).
- ¹⁵¹P. Heikkilä, T. Hirvikorpi, H. Hilden, J. Sievänen, L. Hyvärinen, A. Harlin, and M. Vähä-Nissi, *J. Mater. Sci.* **47**, 3607 (2012).
- ¹⁵²M. Mousa and G. N. Parsons, *ACS Appl. Mater. Interfaces* **7**, 19523 (2015).
- ¹⁵³X. Meng, X.-Q. Yang, and X. Sun, *Adv. Mater.* **24**, 3589 (2012).
- ¹⁵⁴K. H. Seng, J. Liu, Z. P. Guo, Z. X. Chen, D. Jia, and H. K. Liu, *Electrochem. Commun.* **13**, 383 (2011).
- ¹⁵⁵X. Han, Y. Liu, Z. Jia, Y.-C. Chen, J. Wan, N. Weadock, K. J. Gaskell, T. Li, and L. Hu, *Nano Lett.* **14**, 139 (2014).
- ¹⁵⁶X. Chen, E. Pomerantseva, P. Banerjee, K. Gregorczyk, R. Ghodssi, and G. Rubloff, *Chem. Mater.* **24**, 1255 (2012).
- ¹⁵⁷J. S. Daubert, N. P. Lewis, H. N. Gotsch, J. Z. Mundy, D. N. Monroe, E. C. Dickey, M. Losego, and G. N. Parsons, *Chem. Mater.* **27**, 6524 (2015).
- ¹⁵⁸M. I. Mejia, J. M. Marin, G. Restrepo, C. Pulgarin, E. Mielczarski, J. Mielczarski, Y. Arroyo, J.-C. Lavanchy, and J. Kiwi, *Appl. Catal. B-Environ.* **91**, 481 (2009).
- ¹⁵⁹Q. Du, J. Wu, and H. Yang, *ACS Catal.* **4**, 144 (2014).
- ¹⁶⁰M. Kemell, V. Pore, M. Ritala, and M. Leskelä, *Chem. Vap. Deposition* **12**, 419 (2006).
- ¹⁶¹F. Kayaci, S. Vempati, C. Ozgit-Akgun, I. Donmez, N. Biyikli, and T. Uyar, *Nanoscale* **6**, 5735 (2014).
- ¹⁶²C. Liu, C.-C. Wang, C.-C. Kei, Y.-C. Hsueh, and T.-P. Perng, *Small* **5**, 1535 (2009).
- ¹⁶³A. A. Dameron *et al.*, *Appl. Surf. Sci.* **258**, 5212 (2012).
- ¹⁶⁴S. T. Meek, J. A. Greathouse, and M. D. Allendorf, *Adv. Mater.* **23**, 249 (2011).
- ¹⁶⁵T. Suntola and J. Antson, U.S. patent 4,058,430 (15 November 1977).
- ¹⁶⁶P. Poodt, A. Lankhorst, F. Roozeboom, K. Spee, D. Maas, and A. Vermeer, *Adv. Mater.* **22**, 3564 (2010).
- ¹⁶⁷P. Poodt, R. Knaapen, A. Illiberi, F. Roozeboom, and A. van Asten, *J. Vac. Sci. Technol. A* **30**, 01A142 (2012).
- ¹⁶⁸P. Poodt, D. C. Cameron, E. Dickey, S. M. George, V. Kuznetsov, G. N. Parsons, F. Roozeboom, G. Sundaram, and A. Vermeer, *J. Vac. Sci. Technol. Vac. Surf. Films* **30**, 010802 (2012).
- ¹⁶⁹B. Maze, H. Vahedi Tafreshi, Q. Wang, and B. Pourdeyhi, *J. Aerosol Sci.* **38**, 550 (2007).
- ¹⁷⁰T. Tynell, I. Terasaki, H. Yamauchi, and M. Karppinen, *J. Mater. Chem. A* **1**, 13619 (2013).
- ¹⁷¹B. R. Sutherland *et al.*, *Adv. Mater.* **27**, 53 (2015).
- ¹⁷²J. P. Klesko, C. M. Thrush, and C. H. Winter, *Chem. Mater.* **27**, 4918 (2015).
- ¹⁷³J. Slade, M. Agpaoa-Kraus, J. Bowman, A. Riecker, T. Tiano, C. Carey, and P. Wilson, *MRS Proc.* **736**, D3.1.1–D3.1.7 (2002).